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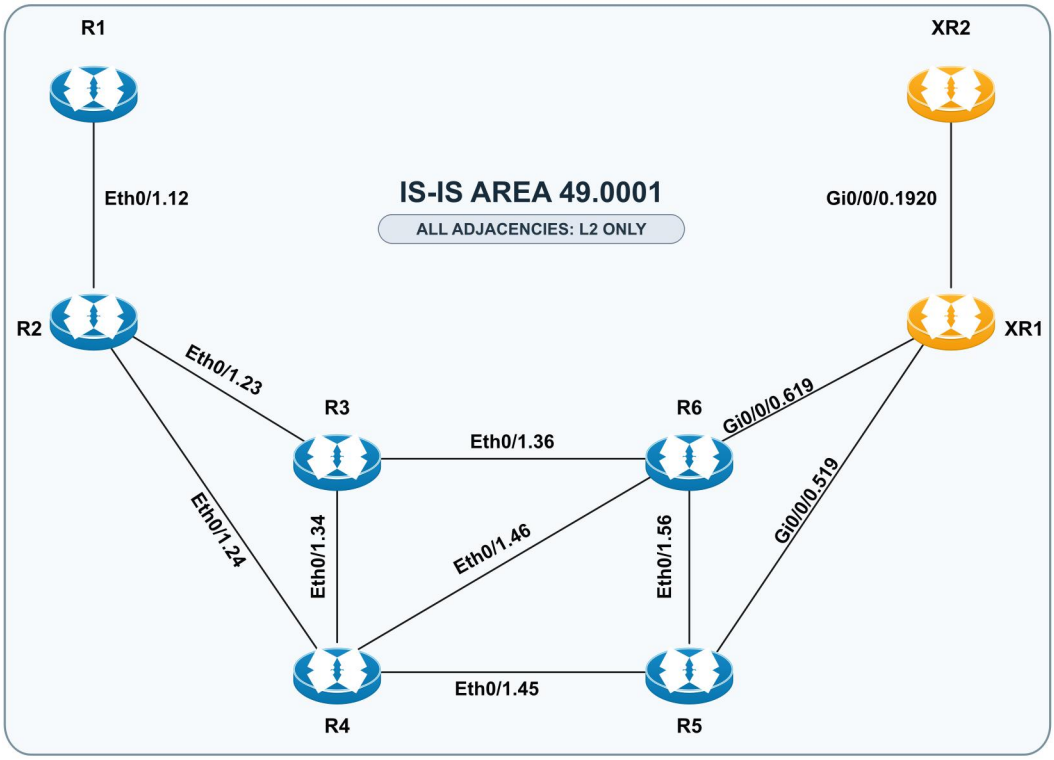
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ISIS

Prefix Suppression



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

- Configure ISIS so that **only** the **loopbacks** are advertised into the **IGP**.
- Only use **interface-level** commands on **R2** and **XR2**.

Answer

```
# R1 - R6
router isis
 advertise passive-only

# R2
interface Ethernet0/1.12
 no isis advertise prefix
interface Ethernet0/1.23
 no isis advertise prefix
```

```
interface Ethernet0/1.24
 no isis advertise prefix

# XR1
router isis 1
 address-family ipv4 unicast
  advertise passive-only

# XR2
router isis 1
 interface GigabitEthernet0/0/0/0.1920
  suppressed
```

Explanation

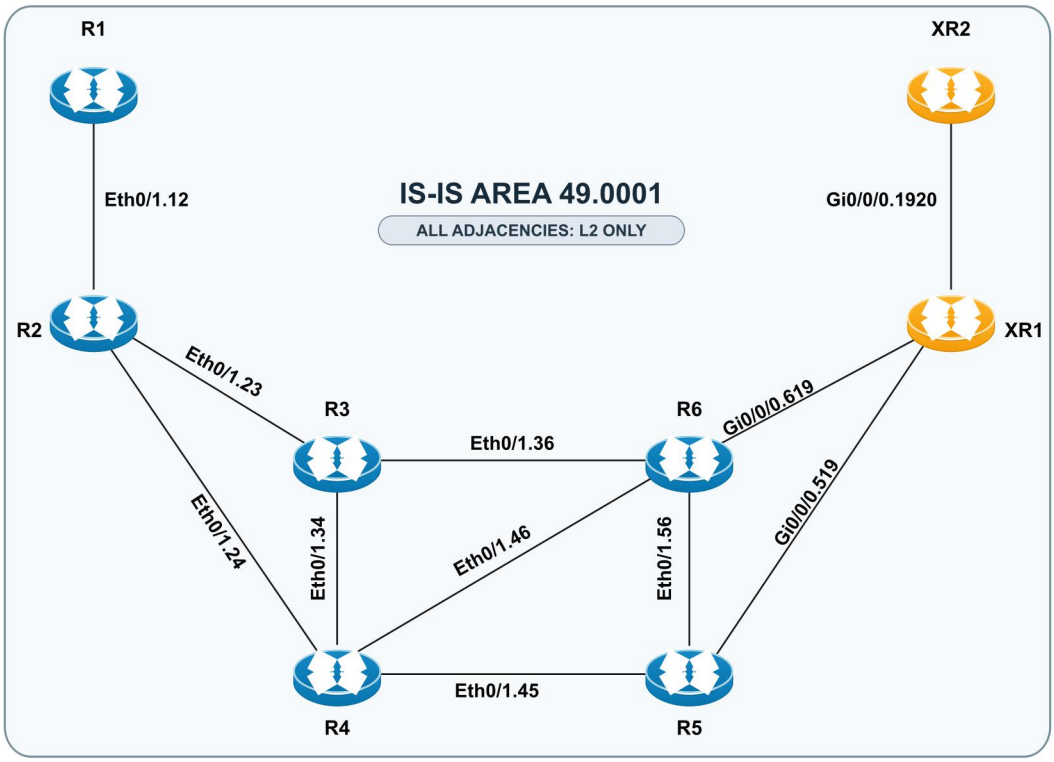
In ISIS, the link address is not part of the underlying topology. It is simply advertised as an additional prefix in the IP reachability TLV. By configuring **advertise passive-only** we instruct the router to suppress all prefixes except for the passive interfaces, which in our case is only Lo0.

```
R1#show isis database R1.00-00 verbose


IS-IS Level-2 LSP R1.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x00000003   0x3E4E                929/*         0/0/0
  Area Address: 49.0001
  NLPID:        0xCC
  Hostname: R1
  Metric: 10         IS R2.03
  IP Address:   1.1.1.1
  Metric: 0          IP 1.1.1.1 255.255.255.255
```

How does R2 have the correct nexthop for R1's loopback? We don't see R1 advertising its nexthop anywhere in the LSP. The answer is that R1 advertises its interface address in the Hello. This is how R2 knows what to use for the nexthop to 1.1.1.1.

Hello Padding



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

- Configure **R1** and **XR2** so that Hellos are only padded the first six times.

Answer

```
# R1
int Eth0/1.12
 no isis hello padding
! Or
router isis
 no hello padding [multi-point|point-to-point]

# XR2
router isis 1
 int Gi0/0/0/0.1920
  hello-padding sometimes [level 2]
```

Explanation

By default, the router will always pad Hellos. This can be considered an unnecessary waste of bandwidth, as once you've verified MTU via the padded Hellos, you no longer really need to continue to pad Hellos.

When using the above configuration, the router will only pad the first 6 Hellos. After this, all subsequent Hellos are not padded, which saves some bandwidth.

If for some reason you wanted to never pad Hellos, not even at the beginning of the adjacency, you can use **no isis hello padding always** under the interface on IOS-XE, or **hello-padding disable** on IOS-XR. This is similar to OSPF's **ip ospf mtu-ignore** setting.

```
R1#show clns interface eth0/1.12
Ethernet0/1.12 is up, line protocol is up
 Attached to: isis
  Checksums enabled, MTU 1497, Encapsulation SAP
  ERPDUs enabled, min. interval 10 msec.
  CLNS fast switching enabled
  CLNS SSE switching disabled
  DEC compatibility mode OFF for this interface
  Next ESH/ISH in 21 seconds
  Routing Protocol: IS-IS
    Circuit Type: level-1-2
    Interface number 0x0, local circuit ID 0x1
    Level-2 Metric: 10, Priority: 64, Circuit ID: R2.03
    DR ID: R2.03
    Level-2 IPv6 Metric: 10
    Number of active level-2 adjacencies: 1
    Next IS-IS LAN Level-2 Hello in 327 milliseconds
    No hello padding
```

When you use **no isis hello padding always**, you will instead see "**Never pad hello PDU**":

```
R1#show clns interface eth0/1.12
Ethernet0/1.12 is up, line protocol is up
 Attached to: isis
  Checksums enabled, MTU 1497, Encapsulation SAP
  ERPDUs enabled, min. interval 10 msec.
  CLNS fast switching enabled
  CLNS SSE switching disabled
  DEC compatibility mode OFF for this interface
  Next ESH/ISH in 39 seconds
  Routing Protocol: IS-IS
    Circuit Type: level-1-2
    Interface number 0x0, local circuit ID 0x1
    Level-2 Metric: 10, Priority: 64, Circuit ID: R2.03
    DR ID: R2.03
    Level-2 IPv6 Metric: 10
    Number of active level-2 adjacencies: 1
    Next IS-IS LAN Level-2 Hello in 8 seconds
    Never pad hello PDU
```

On **IOS-XR**.

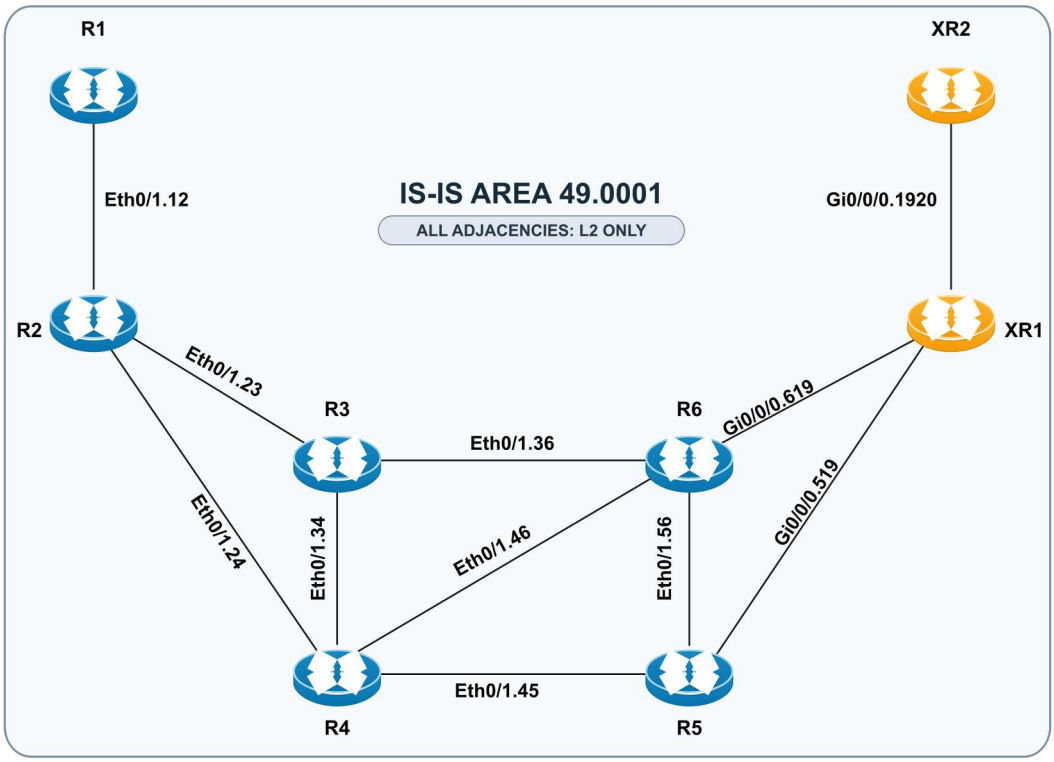
```
RP/0/RP0/CPU0:XR2#show isis interface gigabitEthernet 0/0/0/0.1920
Thu Oct 22 01:51:22.590 UTC

GigabitEthernet0/0/0/0.1920 Enabled
  Adjacency Formation:      Enabled
  Prefix Advertisement:     Enabled
  Bandwidth:                1000000
  RSI SRLG:                 Not Registered

  Circuit Type:             level-2-only (Configured: level-1-2)
  Media Type:               LAN
  Circuit Number:           7
  Last IIH Received:        01:51:22 (0.31 sec ago), 1497 octets
  Last PDU Rejected:        01:16:01 (00:35:20 ago), 101 octets, L2 LSP, Wrong Adjacency
State
  Last IIH Sent:            01:51:20 (2.34 sec ago), 55 octets
```

XR2 sent Hello **without Padding** (55 octets), whilst its neighbor (**XR1**) sent padded Hello (1497 octets).

Overload Bit



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

Set the overload bit on routers R2, R3, and XR1:

1. On **R2**, set overload bit on startup, wait for BGP, and supress all external routes
2. On **R3**, set overload bit right now, and advertise all prefixes
3. On **XR1**, set overload bit on startup, wait for BGP, and suppress both inter-area and external routes

Answer

```
# R2
router isis
```

```
 set-overload-bit on-startup wait-for-bgp suppress external

# R3
router isis
 set-overload-bit

# XR1
router isis 1
 set-overload-bit on-startup wait-for-bgp
```

Explanation

XR1, we can verify overload bit configuration using **show isis protocol**.

```
RP/0/RP0/CPU0:XR1#show isis protocol | utility egrep -A9 Overload
Sun Nov  1 07:12:42.212 UTC
  Overload Bit Level-2: configured, clear
    interarea prefixes suppressed, external prefixes suppressed
      on startup until BGP converges
    clear since Nov 01 2026 07:12:13.366 UTC
    IPv4 Unicast
      clear
  Maximum Metric: not configured
  Topologies supported by IS-IS:
    IPv4 Unicast
      Rib connected
```

R3 is currently setting the OL bit in its LSP:

```
R2#show isis database

IS-IS Level-2 Link State Database:
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00              0x00000002   0x2BBF                901/1196      0/0/0
R2.00-00            * 0x00000004   0x8CBC                899/*         0/0/0
R2.01-00            * 0x00000001   0x0ECB                900/*         0/0/0
R3.00-00              0x00000004   0x9490                982/1200      0/0/1
R3.01-00              0x00000001   0x2DA9                899/1199      0/0/0
R4.00-00              0x00000003   0x2329                901/1198      0/0/0
R4.01-00              0x00000001   0x33A1                898/1199      0/0/0
```

Note that when the OL bit is set, the router is ***ị ḅúḅó*used for transit**, even if no other paths exist. The node can still be used to reach leaf prefixes advertised directly from the node itself, such as its own loopback.

R2 should no longer have ECMP routes to XR1. It should only be using R4 for transit. It will only use R3 for traffic R3 itself is advertising - its loopback and connected routes.

```
R2#show ip route 20.20.20.20
Routing entry for 20.20.20.20/32
  Known via "isis", distance 115, metric 40, type level-2
  Redistributing via isis
  Last update from 20.2.4.4 on Ethernet0/1.24, 00:05:48 ago
  Routing Descriptor Blocks:
  * 20.2.4.4, from 20.20.20.20, 00:05:48 ago, via Ethernet0/1.24
      Route metric is 40, traffic share count is 1
```

```
R6#siib
Interface              IP-Address      OK? Method Status                Protocol
Ethernet0/0            10.200.255.7    YES TFTP   up                    up
Ethernet0/1.36         20.3.6.6        YES TFTP   administratively down down
Ethernet0/1.46         20.4.6.6        YES TFTP   administratively down down
Ethernet0/1.56         20.5.6.6        YES TFTP   administratively down down
Ethernet0/1.619        20.6.19.6       YES TFTP   up                    up
Loopback0              6.6.6.6         YES TFTP   up                    up

Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
  1 20.6.19.19 1 msec 0 msec 1 msec
  2 20.5.19.5 0 msec 0 msec 1 msec
  3 20.4.5.4 0 msec 0 msec 0 msec
  4 20.2.4.2 1 msec 0 msec 1 msec
```

LSP Size

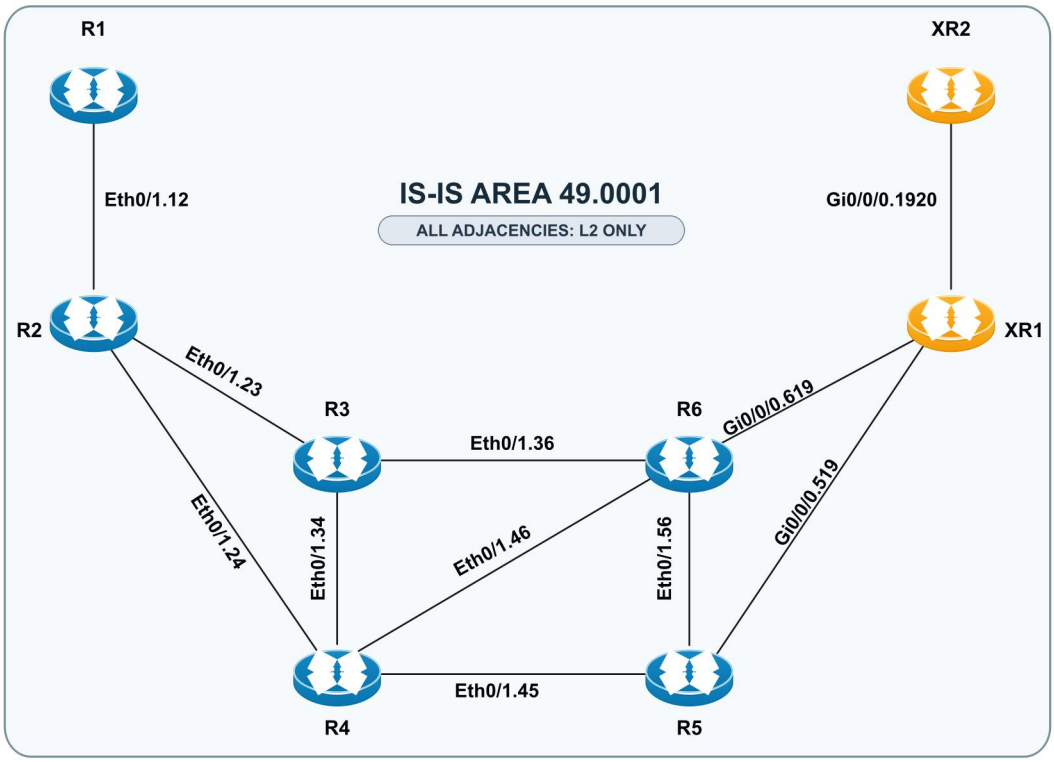


Figure 0-1 ISIS Single Area

Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

1. Enable wide metrics everywhere.
2. Configure R1 and XR2 so that their **LSP** is **fragmented** once is it bigger than **128 bytes**.

Answer

```
# IOL
router isis
 metric-style wide

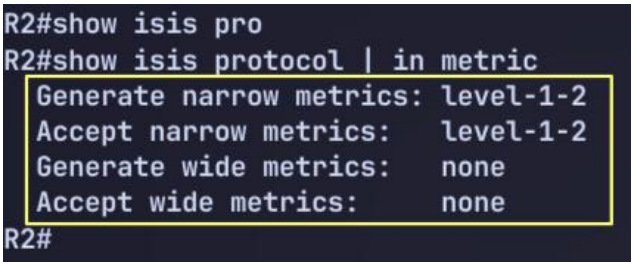
# IOS-XR
```

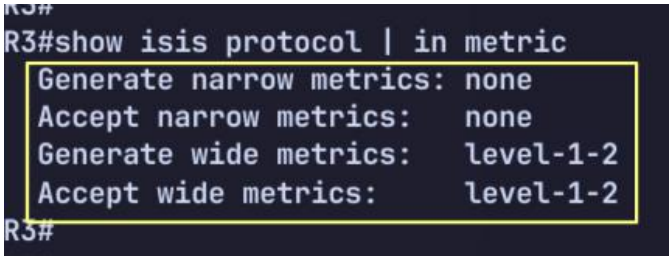
```
router isis 1
 address-family ipv4 unicast
  metric-style wide

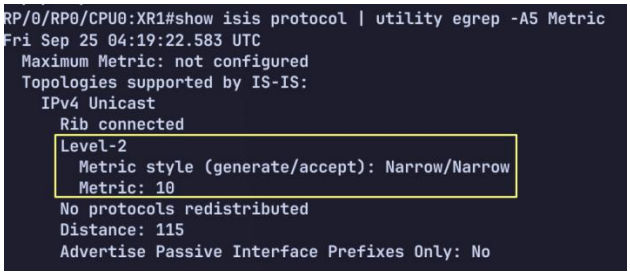
# R1
router isis
 lsp-mtu 128

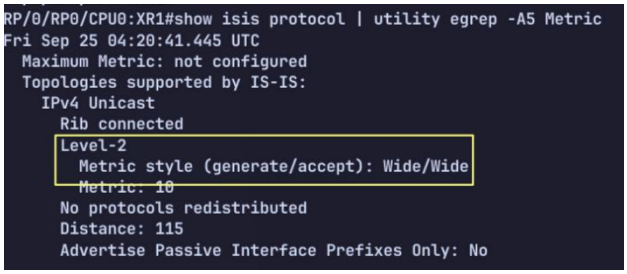
# XR2
router isis 1
 lsp-mtu 128
```

Explanation









By default, an LSP is fragmented at 1500 bytes. We can control this by setting the lsp-mtu. When the router's own LSP becomes larger than the specified MTU size, it is split into multiple LSPs.

Default Metric

Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
```

```
commit replace confirmed minutes 1
commit
```

1. Configure all routers so that the default metric is 100.

Answer

```
# IOL
router isis
 metric-style wide
 metric 100

# IOS-XR
router isis 1
 address-family ipv4 unicast
  metric-style wide
  metric 100
```

Explanation

```
# R1
R1#show isis topology


IS-IS TID 0 paths to level-2 routers
System Id            Metric     Next-Hop             Interface   SNPA
R1                   --
R2                   100        R2                   Et0/1.12    aabb.cc00.0210
R3                   200        R2                   Et0/1.12    aabb.cc00.0210
R4                   200        R2                   Et0/1.12    aabb.cc00.0210
R5                   300        R2                   Et0/1.12    aabb.cc00.0210
R6                   300        R2                   Et0/1.12    aabb.cc00.0210
XR1                  400        R2                   Et0/1.12    aabb.cc00.0210
XR2                  500        R2                   Et0/1.12    aabb.cc00.0210
```

To use a default **metric of 100**, we must use **wide metrics**. First we enable wide metrics everywhere.

The **metric** command under the ISIS router process dictates the default metric for all interfaces. Alternatively, you can manually configure the metric under each interface individually.

Hello/Hold Timer

Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

1. Change the **R1-R2** link to be **p2p**. Configure the **Hello/Hold** timers to be **2/6** seconds.
2. Change the **hello/hold** timer on **R3** so that its **Eth0/1.23** interface uses a hold timer of **3 seconds** when it is elected DIS on the LAN.

Answer

```
# R1 & R2
interface Ethernet0/1.12
 isis network point-to-point
 isis hello-multiplier 3
 isis hello-interval 2

# R3
interface Ethernet0/1.23
 isis hello-interval 3
 isis hello-multiplier 3
```

Note : the **hello-multiplier** command was not strictly necessary above. This is because the default value is 3.

Explanation

Before configuring isis network type

```
R1#show clns interface eth0/1.12
Ethernet0/1.12 is up, line protocol is up
 Attached to: isis
  Checksums enabled, MTU 1497, Encapsulation SAP
  ERPDUs enabled, min. interval 10 msec.
  CLNS fast switching enabled
  CLNS SSE switching disabled
  DEC compatibility mode OFF for this interface
  Next ESH/ISH in 18 seconds
  Routing Protocol: IS-IS
    Circuit Type: level-1-2
    Interface number 0x0, local circuit ID 0x1
    Level-2 Metric: 100, Priority: 64, Circuit ID: R2.01
    DR ID: R2.01
    Level-2 IPv6 Metric: 10
    Number of active level-2 adjacencies: 1
    Next IS-IS LAN Level-2 Hello in 2 seconds
```

```
R1#show clns interface eth0/1.12
Ethernet0/1.12 is up, line protocol is up
 Attached to: isis
  Checksums enabled, MTU 1497, Encapsulation SAP
  ERPDUs enabled, min. interval 10 msec.
  CLNS fast switching enabled
  CLNS SSE switching disabled
  DEC compatibility mode OFF for this interface
  Next ESH/ISH in 13 seconds
  Routing Protocol: IS-IS
    Circuit Type: level-1-2
    Interface number 0x0, local circuit ID 0x7
    Neighbor Extended Local Circuit ID: 0x7
    Neighbor System-ID: R2
    P2P retransmit queue size: 0
    P2P retransmits: 0
    LSP fast flooding: Disabled
```

Verify holdtime. This is the **Neighbor's Holdtime**, not local timer.

```
R1#show isis nei

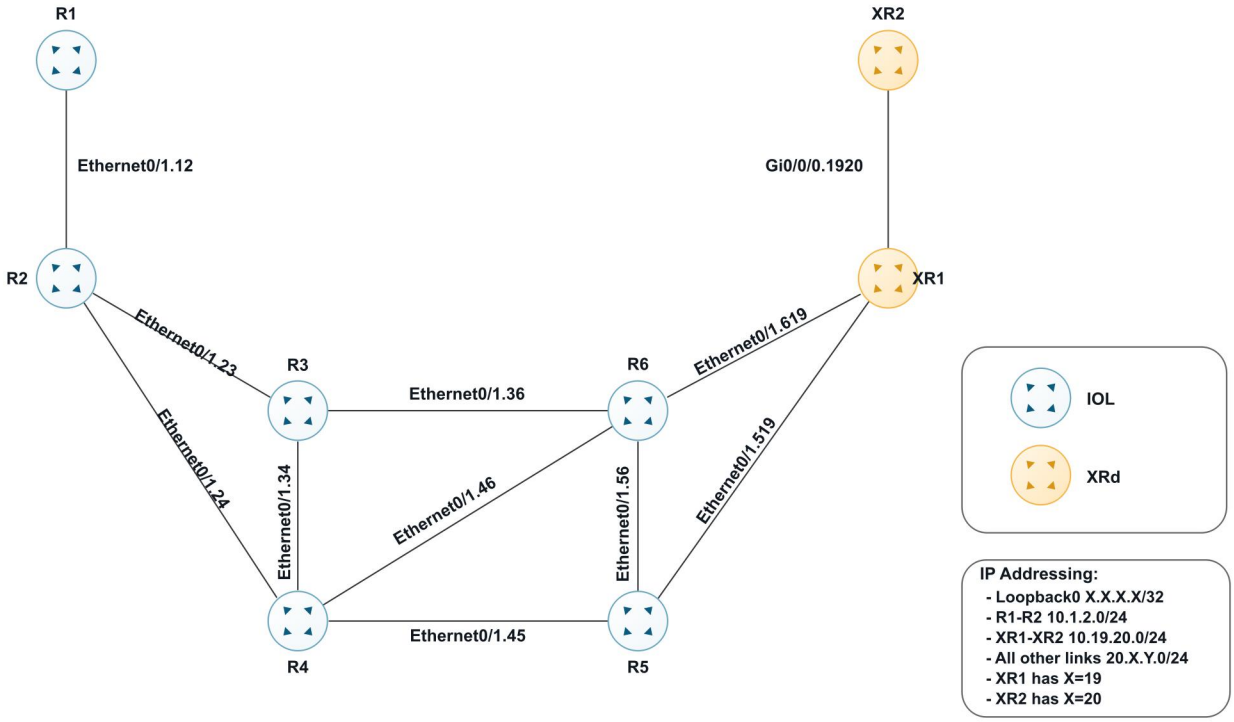
System Id       Type Interface     IP Address      State Holdtime Circuit Id
R2              L2   Et0/1.12      10.1.2.2        UP    5        07
R1#show isis nei

System Id       Type Interface     IP Address      State Holdtime Circuit Id
R2              L2   Et0/1.12      10.1.2.2        UP    4        07
R1#
```

We can verify local timer on IOS-XR.

```
RP/0/RP0/CPU0:XR1#show isis interface gigabitEthernet 0/0/0/0.1920 | include "Hello"
Thu Sep 24 07:12:22.369 UTC
    Hello Interval:         3 s
    Hello Multiplier:       5
```

Mesh Groups



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

Configure **R2** so that LSPs received on **eth0/1.12** are only forwarded out **eth0/1.24**. LSPs received on **eth0/1.23** should only be flooded out **eth0/1.24** as well.

Answer

```
# R2

interface Ethernet0/1.12
 isis mesh-group 10

interface Ethernet0/1.23
 isis mesh-group 10
```

Explanation

By default, an **LSP received** on any interface will be **flooded out all other interfaces**. This can be an inefficient use of bandwidth in large scale, highly meshed topologies. To better scale ISIS, we can configure interfaces into groups for flooding.

If an interface belongs to an ISIS mesh-group, any LSP received on that interface is **only flooded** out other interfaces **that are NOT in the same mesh group**.

```
LSP recv on .12
   ├──> .23     same mesh-group
   └──> .24

LSP recv on .23
   ├──> .12     same mesh-group
   └──> .24
```

Before Mesh Group

Use **`debug isis update-packet`** on R2, and Shutdown **lo0** on **R1** to force R1 to update its LSP.

```
ISIS-Upd: from SNPA aabb.cc00.0110 (Ethernet0/1.12)
Upd ISIS-Upd: LSP newer than database copy
Upd ISIS-Upd: TLV contents different, code 132
Upd ISIS-Upd: TID 0 leaf routes changed
Upd ISIS-Upd: Sending L2 LSP 0000.0000.0001.00-00, seq 6
Upd ISIS-Upd: ht 1199 on Ethernet0/1.24 via high priority queue
Upd ISIS-Upd: Sending L2 LSP 0000.0000.0001.00-00, seq 6
Upd ISIS-Upd: ht 1199 on Ethernet0/1.23 via high priority queue
Dec ISIS-Upd: Node cache: processing R1.00-00 Level 2
Dec ISIS-Upd: Node cache: lsp_zero found R1.00-00 Level 2
Dec ISIS-Upd: Node cache Parse lsp R1.00-00 Level 2
Dec ISIS-Upd: Node cache update: Node found: yes lsp R1.00-00 Level 2
```

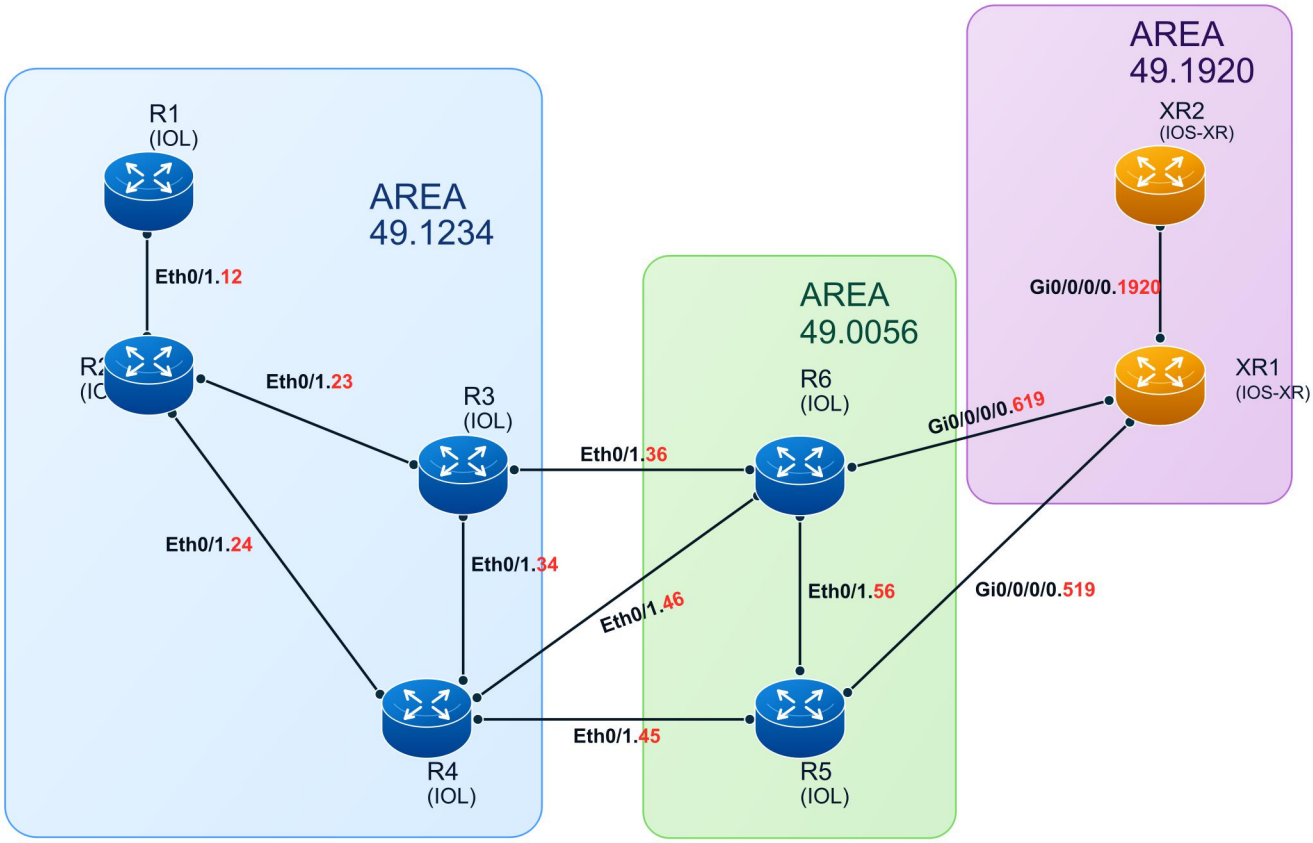
After Mesh Group

LSP received on **eth0/1.12** is flooded only to **eth0/1.24**.

```
ISIS-Upd: Rec L2 LSP 0000.0000.0001.00-00, seq 7, ht 1200,
ISIS-Upd: from SNPA aabb.cc00.0110 (Ethernet0/1.12)
Upd ISIS-Upd: LSP newer than database copy
Upd ISIS-Upd: TLV contents different, code 132
Upd ISIS-Upd: TID 0 leaf routes changed
Upd ISIS-Upd: Sending L2 LSP 0000.0000.0001.00-00, seq 7
Upd ISIS-Upd: ht 1199 on Ethernet0/1.24 via high priority queue
Dec ISIS-Upd: Node cache: processing R1.00-00 Level 2
Dec ISIS-Upd: Node cache: lsp_zero found R1.00-00 Level 2
```

```
Dec ISIS-Upd: Node cache Parse lsp R1.00-00 Level 2
Dec ISIS-Upd: Node cache update: Node found: yes lsp R1.00-00 Level 2
```

Prefix Summarization



Load Config

```
# IOL
configure replace unix:labconfigs/isis.inter.area.cfg force

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace confirmed minutes 1
commit
```

The topology has been changed so that there are three ISIS areas now.

- Configure **R3** and **R4** to summarize **R1**'s **loopback** as 1.1.1.0/24 into **Level 2**.
- Configure **R4** to summarize **R5**'s loopback as **5.5.5.0/24** into **Level 1**. No other prefixes should be leaked into Level 1.
- Configure **XR1** to summarize **XR2**'s loopback as 20.20.20.0/24 into **Level 2**.
- Configure **XR2** to redistribute a new prefix, 20.20.200.20/32 into ISIS and **summarize it** as 20.20.200.0/24.

Answer

```
# R3 & R4
router isis
  Summary-address 1.1.1.0 255.255.255.0 level-2

# R4

ip prefix-list L2_To_L1 seq 5 permit 5.5.5.5/32

route-map L2_To_L1 permit 0
 match ip address prefix-list L2_To_L1

router isis
 summary-address 5.5.5.0 255.255.255.0 level-1
 redistribute isis ip level-2 into level-1 route-map L2_To_L1

# XR1
router isis 1
 address-family ipv4 unicast
  summary-prefix 20.20.20.0/24 level 2

# XR2
interface Loopback20
 ipv4 address 20.20.200.20 255.255.255.255
!
route-policy Lo20
  if destination in (20.20.200.20/32) then
    pass
  endif
end-policy
!
router isis 1
 address-family ipv4 unicast
  summary-prefix 20.20.200.0/24
  redistribute connected level-1 route-policy Lo20
 !
!
end
```

R1's Loopback Before / After

```
R6#show ip route 1.1.1.1
Routing entry for 1.1.1.1/32
  Known via "isis", distance 115, metric 30, type level-2
  Redistributing via isis
  Last update from 20.3.6.3 on Ethernet0/1.36, 00:05:51 ago
  Routing Descriptor Blocks:
  * 20.4.6.4, from 4.4.4.4, 00:05:51 ago, via Ethernet0/1.46
      Route metric is 30, traffic share count is 1
    20.3.6.3, from 3.3.3.3, 00:05:51 ago, via Ethernet0/1.36
      Route metric is 30, traffic share count is 1


R6#show ip route 1.1.1.1
Routing entry for 1.1.1.0/24
  Known via "isis", distance 115, metric 30, type level-2
  Redistributing via isis
  Last update from 20.4.6.4 on Ethernet0/1.46, 00:00:13 ago
```

```
  Routing Descriptor Blocks:
    20.4.6.4, from 4.4.4.4, 00:00:13 ago, via Ethernet0/1.46
      Route metric is 30, traffic share count is 1
  * 20.3.6.3, from 3.3.3.3, 00:00:13 ago, via Ethernet0/1.36
      Route metric is 30, traffic share count is 1
```

R5's Loopback Before / After

```
R2#show ip route 5.5.5.5
% Network not in table

R2#show ip route 5.5.5.5
Routing entry for 5.5.5.0/24
  Known via "isis", distance 115, metric 20, type inter area
  Redistributing via isis
  Last update from 20.2.4.4 on Ethernet0/1.24, 00:00:48 ago
  Routing Descriptor Blocks:
  * 20.2.4.4, from 4.4.4.4, 00:00:48 ago, via Ethernet0/1.24
      Route metric is 20, traffic share count is 1
```

XR2's Loopback

```
R6#show ip route 20.20.20.20
Routing entry for 20.20.20.20/32
  Known via "isis", distance 115, metric 20, type level-2
  Redistributing via isis
  Last update from 20.6.19.19 on Ethernet0/1.619, 00:13:59 ago
  Routing Descriptor Blocks:
  * 20.6.19.19, from 19.19.19.19, 00:13:59 ago, via Ethernet0/1.619
      Route metric is 20, traffic share count is 1


R6#show ip route 20.20.20.20
Routing entry for 20.20.20.0/24
  Known via "isis", distance 115, metric 20, type level-2
  Redistributing via isis
  Last update from 20.6.19.19 on Ethernet0/1.619, 00:00:15 ago
  Routing Descriptor Blocks:
  * 20.6.19.19, from 19.19.19.19, 00:00:15 ago, via Ethernet0/1.619
      Route metric is 20, traffic share count is 1
```

Explanation

When summarizing level 1 prefixes as they are advertised into level 2, you use the command **summary-address <prefix> <mask> level-2**. Optionally, you can leave the **level-2** keyword out, as level 2 is the default.

When summarizing level 2 prefixes as they are advertised into level 1, you must first ensure that the routes are being leaked into level 1 in the first place. Then you use the summary command and specify **level-1**. The level is always the level you are advertising the prefixes into.

Default Route Preference

Load Config

```
# IOL
configure replace unix:labconfigs/isis.inter.area.cfg force

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace confirmed minutes 1
commit
```

- Configure area **49.1234** so that **R2** prefers **R3** for the **default route**, and only uses **R4** upon loss of R3. **Do not** use **metrics** to achieve this.

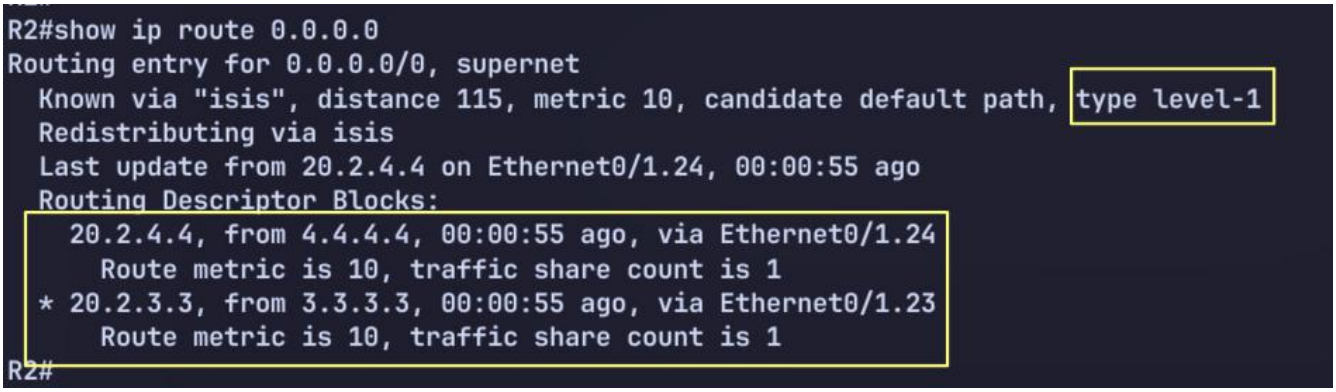
Answer

```
router isis
 default-information originate route-map SET_L1
route-map SET_L1 permit 10
 set level level-1
```

Explanation

ISIS allows us to **originate a default route** just like OSPF. An L1 router will **prefer an explicit 0/0 route** that is advertised in the area **ove**r using the **ATT bit**.

Before Config



```
R2#show isis database R3.00-00 verbose


IS-IS Level-1 LSP R3.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R3.00-00              0x00000029   0x83D3                1195/1200      1/0/0
  Area Address: 49.1234
  NLPID:        0xCC
  Hostname: R3
  Metric: 10         IS R2.00
  IP Address:   3.3.3.3
  Metric: 0          IP 3.3.3.3 255.255.255.255
  Metric: 10         IP 20.2.3.0 255.255.255.0
R2#
```

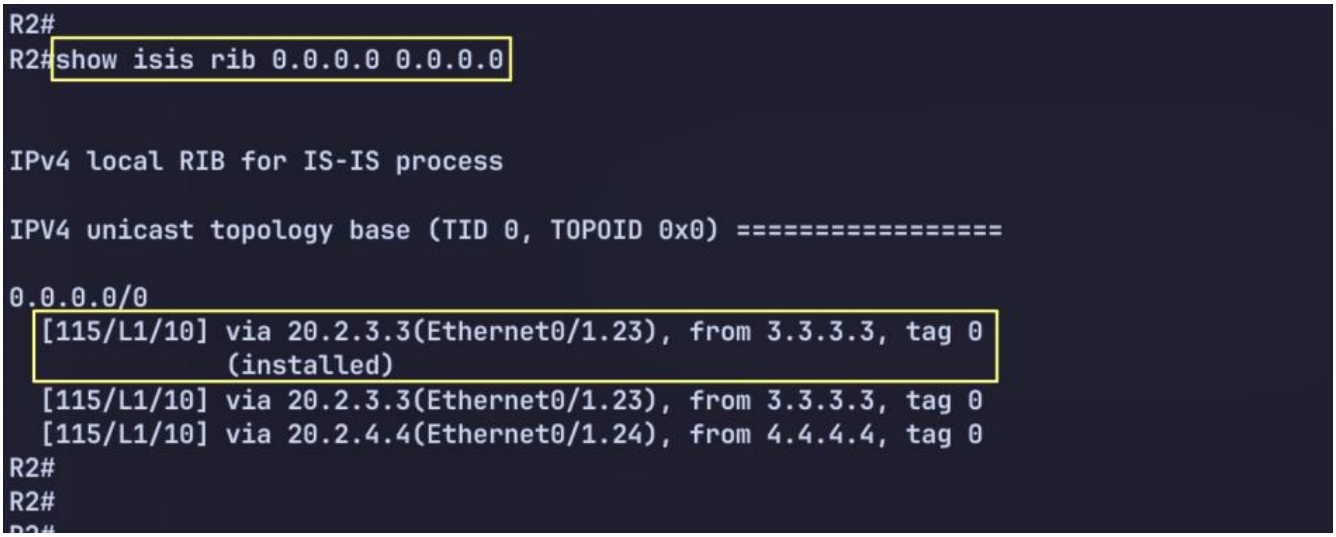
After Configuring default-information originate

```
R2#
R2#show ip route 0.0.0.0
Routing entry for 0.0.0.0/0, supernet
  Known via "isis", distance 115, metric 10, candidate default path, type level-1
  Redistributing via isis
  Last update from 20.2.3.3 on Ethernet0/1.23, 00:03:41 ago
  Routing Descriptor Blocks:
  * 20.2.3.3, from 3.3.3.3, 00:03:41 ago, via Ethernet0/1.23
      Route metric is 10, traffic share count is 1
R2#
```

```
R2#show isis database R3.00-00 verbose


IS-IS Level-1 LSP R3.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R3.00-00              0x0000002A   0x12B6                1038/1200      1/0/0
  Area Address: 49.1234
  NLPID:        0xCC
  Hostname: R3
  Metric: 10         IS R2.00
  IP Address:   3.3.3.3
  Metric: 0          IP 3.3.3.3 255.255.255.255
  Metric: 10         IP 20.2.3.0 255.255.255.0
  Metric: 0          IP 0.0.0.0 0.0.0.0
R2#
```

ISIS RIB still has both 0/0 routes generated from the ATT bit set on R3 and R4, but the explicit 0/0 is preferred and installed.



ISIS Timers

Load Config

```
# IOL
configure replace unix:labconfigs/isis.inter.area.cfg force

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace confirmed minutes 1
commit
```

Set **R1** and **XR2** to use the following ISIS timers:

- **Local LSP**s should have a **max lifetime** of 600 seconds. These LSPs should be **refreshed** every 300 seconds.
- Locally-generated LSPs should be throttled **500 msec** after **a change**, **750 msec** if occuring after an LSP was **just generated** previously, and a **max backoff** of 3 seconds.
- **SPF calculation** should be throttled **250 msec** after receiving an update, **500 msec** if an SPF run has **just occured**, and a **max wait** time of 3 seconds.
- **Partial route calculation** should use the same timers as SPF calculation.
- During flooding, interfaces should pace LSPs at 20 msec intervals
- During flooding, the router should allow for up to 3 seconds before retransmitting the same LSP. If LSPs do need to be retransmitted, these should be paced at 40 msec intervals.

Answer

```
# R1
router isis
 max-lsp-lifetime 600
 lsp-refresh-interval 300
 spf-interval 3 250 500
 prc-interval 3 250 500
 lsp-gen-interval 3 500 750
!
int Eth0/1.12
 isis lsp-interval 20
 isis retransmit-interval 3
 isis retransmit-throttle-interval 40

# XR2
router isis 1
 max-lsp-lifetime 600
 lsp-refresh-interval 300
 lsp-gen-interval initial-wait 500 secondary-wait 750 maximum-weight 3000
 !
 address-family ipv4 unicast
  spf-interval initial-wait 250 second-wait 500 maximum-wait 3000
 !
 int Gi0/0/0/0.1920
  lsp-interval 20
  retransmit-interval 3
  retransmit-throttle-interval 40
```

Explanation

The **max-lsp-lifetime** command sets the lifetime on locally generated LSPs. By default this appears to be 1200 seconds. This defines how long an LSP is valid in the LSDB. A longer interval allows for less frequent refreshes, which saves some bandwidth in a large network.

Log Neighbor Changes

Load Config

```
# IOL
configure replace unix:labconfigs/isis.inter.area.cfg force

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace confirmed minutes 1
commit
```

Configure **R1** so that when **eth0/1.12** is manually shut down, the following log message appears:

```
%CLNS-5-ADJCHANGE: ISIS: Adjacency to R2 (Ethernet0/1.12) Down, interface
deleted(non-iih)
```

Configure R2 so that when **eth0/1.12** on **R1** is shut down, no log message appears at all.

Answer

```
# R1
router isis
 log-adjacency-changes all

# R2
router isis
 no log-adjacency-changes
```

Explanation

By default, **log-adjacency-changes** is enabled. However, this only logs adjacency changes that are detected through the ISIS Hello. If the interface goes down, the adjacency is lost based on the interface, not the missing Hellos, so no log is generated. If we use the **all** keyword, it will log all adjacency changes, even those caused by non-Hello events.

Troubleshooting 1 – No routes

Load Config

```
# IOL
configure replace unix:labconfigs/isis.tshoot1.cfg

# IOS-XR
configure
load config:labconfigs/isis.tshoot1.cfg
commit replace confirmed minutes 1
commit
```

R1 and XR2 have no ISIS routes in the routing table. Fix the issue without disabling any features that have been enabled on any other routers. You may only configure two routers.

Answer

```
# R1
router isis
 metric-style wide / transition

# XR2
router isis 1
 address-family ipv4 unicast
  metric-style wide / transition
```

Explanation

If we look at the topology on R1 and XR2, we can see that all routers have ** for the metric. This indicates a **metric-style mismatch**:

```
R1#show isis topology


IS-IS TID 0 paths to level-2 routers
System Id            Metric     Next-Hop             Interface   SNPA
R1                   --
R2                   **
R3                   **
R4                   **
R5                   **
R6                   **
XR1                  **
XR2                  **


R2#show isis topology
```

```
IS-IS TID 0 paths to level-2 routers
System Id            Metric     Next-Hop             Interface   SNPA
R1                   **
R2                   --
R3                   10         R3                   Et0/1.23    aabb.cc00.0310
R4                   10         R4                   Et0/1.24    aabb.cc00.0410
R5                   20         R4                   Et0/1.24    aabb.cc00.0410
R6                   20         R3                   Et0/1.23    aabb.cc00.0310
                                R4                   Et0/1.24    aabb.cc00.0410
XR1                  30         R3                   Et0/1.23    aabb.cc00.0310
                                R4                   Et0/1.24    aabb.cc00.0410
XR2                  **

R1#show isis protocol | in metrics
  Generate narrow metrics: level-1-2
  Accept narrow metrics:   level-1-2
  Generate wide metrics:   none
  Accept wide metrics:     none

R2#show isis protocol | in metrics
  Generate narrow metrics: none
  Accept narrow metrics:   none
  Generate wide metrics:   level-1-2
  Accept wide metrics:     level-1-2

RP/0/RP0/CPU0:XR1#show isis protocol | in Metric
Tue Oct 20 04:10:27.157 UTC
  Maximum Metric: not configured
        Metric style (generate/accept): Wide/Wide
        Metric: 10

RP/0/RP0/CPU0:XR2#show isis protocol | in Metric
Tue Oct 20 04:10:27.147 UTC
  Maximum Metric: not configured
        Metric style (generate/accept): Narrow/Narrow
        Metric: 10
```

Troubleshooting 2 – Adjacency

Load Config

```
# R1, R2
configure replace unix:labconfigs/isis.tshoot2.cfg
```

Fix the broken adjacency between **R1** and **R2**.

Answer

Cisco explains that in **IPv6 single-topology IS-IS**, routers must have the **same set of address families** on the **same link**, and IS-IS performs a **consistency check** on the IPv6 information within the hello packets. If one side runs IPv4+IPv6 while the other side only runs IPv4, the adjacency can be rejected or encounter issues.

```
R1                              R2
E0/1.12 ————————————————————— E0/1.12

IPv4 IS-IS                     IPv4 IS-IS
IPv6 IS-IS                     IPv6 IS-IS
IPv6 link-local   ?             IPv6 link-local
```

alternative option for transition

```
# R1

router isis
 address-family ipv6
  no adjacency-check
```

Explanation

There are several things wrong here. The following must match for an adjacency to come up:

- Common subnet for IPv4
- Protocols supported must match and multitopology mode must match
    - However, if one side uses ST and one uses MT transition, then both IPv4 and IPv6 will work
- Circuit types and router levels must match/be compatible
- Hello auth must match
- MTU must match

The following was broken in this situation:

- R2 has a misconfigured key-string
- R2 is in the wrong IPv4 subnet
- R1 does not use p2p as the circuit type
- R1 is level-1 only, while R2 is level-2 only
- R1 has a larger MTU on the interface. R2 won't be able to accept fully padded Hellos from R1
- R1 is running IPv6 on the interface and R2 is not

- **IOS-XE** uses **single-topology** by **default**, in which both IPv4 and IPv6 are required if one side has IPv6 enabled.
- An alternative would be to enable wide metrics on both sides, and then use multi-topology on R1. This would allow the adjacency to come up without R2 supporing IPv6. This is because each protocol has its own topology now on R1.

Troubleshooting 3

Load Config

```
# R1..R6
configure replace unix:labconfigs/isis.tshoot3.cfg
```

All **adjacency** is up, but **R3** to **R6** can't ping **R1**'s loopback using their own loopback as source.

Answer

```
# R1, R2
int eth0/1.12
  no isis circuit-type

# R1
router isis
  is-type level-1-2
```

Explanation

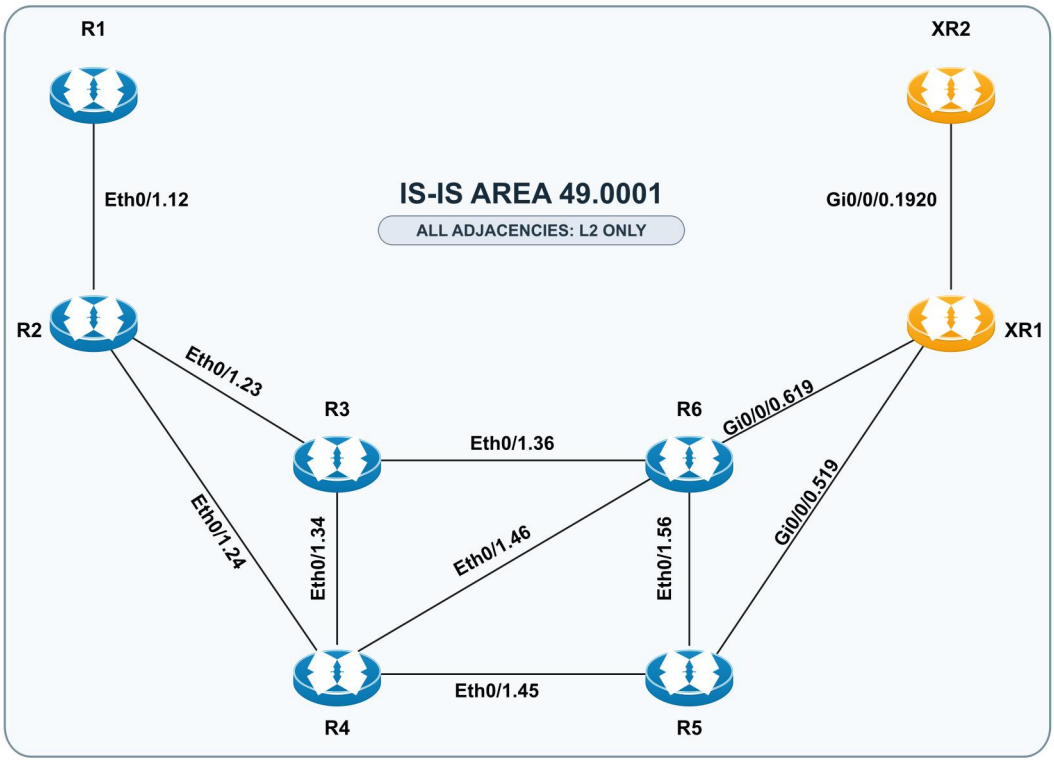
All routers are in the same area, but **R1-R2** adjacency is **L1** only. R1 only know R2, however R2 won't send default route to R1 because it **isn't connected** to **other area**.

```
R1#show clns
Global CLNS Information:
  1 Interfaces Enabled for CLNS
  Configuration Timer: 60, Default Holding Timer: 300, Packet Lifetime 64
  ERPDU's requested on locally generated packets
  Running IS-IS in IP-only mode (CLNS forwarding not allowed)
  NET: 49.0001.0000.0000.0001.00

R1#show isis database

IS-IS Level-1 Link State Database:
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x0000001A   0x2E1F                1197/*         0/0/0
R2.00-00              0x00000010   0x8795                1196/1200      0/0/0
```

IPv6 Single Topology



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.ipv6.init.cfg

# IOS-XR
configure
load config:labconfigs/isis.ipv6.init.cfg
commit replace
y
```

All routers have **IPv6 link-local addresses** enabled on their interfaces, and a loopback IPv6 address of **2001:db8::<router num>/128**.

- **Without** enabling **wide metrics**, configure IPv6 reachability between all routers on their **loopbacks**.

Answer

IOS-XE's default **IPv6 mode** is **single topology**. In single topology, IPv6 "uses" the IPv4 topology. The IPv6 prefixes ride on top of the IPv4 topology and its adjacencies. IPv6 prefixes are simply advertised with TLV 236. This is used whether or not wide metrics are used. (Either can be used with single topology).

The link-local address, used for the nexthop by the neighbor for IPv6 routes, is advertised in the Hello in the IPv6 interface address TLV (232).

On IOS-XR, we **must enable single-topology** under the AFI. By default IOS-XR uses multi topology, in which IPv4 and IPv6 each have a separate topology, and IPv6 is not required to be enabled on every interface.

In summary:

- IPv6 can be used with either single topology or multi topology
    - In single topology, IPv6 uses the IPv4 topology. IPv6 prefixes are advertised in a separate IPv6 reachability TLV (236)
- Single topology is the default mode on IOS-XE
- Multi topology is the default mode on IOS-XR
- In single topology, every link must have both IPv4 and IPv6. If any router does not have IPv6 enabled, both IPv4 and IPv6 reachability is broken.
    - This is because the adjacency requires all protocols be supported between both neighbors
    - The exception is that you can use single topology and **only** IPv6. No IPv4 anywhere in the topology. In this case, the protocols supported TLV lists only IPv6 by itself.
- Single topology can use either narrow or wide metrics. **Multi topology requires wide metrics**.

IPv6 Single Topology Challenge

Load

```
# IOS-XE
config replace unix:labconfigs/isis.st.challenge.init.cfg

# IOS-XR
configure
load config:labconfigs/isis.st.challenge.init.cfg
commit replace
y
```

- R1 and XR2 have enabled IPv6 support using ST mode.
- R2 and XR1 have not yet implemented IPv6.
- The operator has found that the adjacencies between R1-R2 and XR1-XR2 are now broken.
- Find a way to fix this without enabling IPv6 support on R2 or XR1, and without disabling IPv6 support on R1 or XR2.

Answer

```
# R1
router isis
 address-family ipv6
  no adjacency-check

# XR2
router isis 1
 address-family ipv6 unicast
  adjacency-check disable
 !
!
```

Explanation

A router that enables IPv6 on an interface using ST mode requires that the neighbor supports both IPv4 and IPv6 in the protocols supported TLV. If the neighbor only supports IPv4, the router will not bring the adjacency up.

```
R1#show isis database R1.00-00 verbose


IS-IS Level-2 LSP R1.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x00000002   0xEAF2                788/*         0/0/0
  Area Address: 49.0001
  NLPID:        0xCC 0x8E
  Hostname: R1
  IP Address:   1.1.1.1
  Metric: 0          IP 1.1.1.1 255.255.255.255
  Metric: 10         IP 10.1.2.0 255.255.255.0
  IPv6 Address: 2001:DB8::1
  Metric: 0          IPv6 2001:DB8::1/128
    Prefix-attr: X:0 R:0 N:1
R1#
```

```
R2#
R2#show isis database R2.00-00 verbose


IS-IS Level-2 LSP R2.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd
R2.00-00            * 0x0000000A   0xC561                1197/*
  Area Address: 49.0001
  NLPID:        0xCC
  Hostname: R2
  Metric: 10         IS R3.01
  IP Address:   2.2.2.2
  Metric: 0          IP 2.2.2.2 255.255.255.255
  Metric: 10         IP 10.1.2.0 255.255.255.0
  Metric: 10         IP 20.2.3.0 255.255.255.0
R2#
```

To get around this, we can configure the router to disable this adjacency check.

IPv6 Multitopology

Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.ipv6.init.cfg

# IOS-XR
configure
load config:labconfigs/isis.ipv6.init.cfg
commit replace
y
```

All routers have **IPv6 link-local** addresses enabled on their interfaces, and a loopback IPv6 address of 2001:db8::<router num>/128.

Configure **IPv6 reachability** using **multi topology**.

Answer

```
# IOS-XE
router isis
 metric-style wide
 address-family ipv6
  multi-topology
int xxx
 ipv6 router isis

# IOS-XR
router isis 1
 address-family ipv4 unicast
  metric-style wide
 !
 interface Loopback0
  address-family ipv6 unicast
  !
 !
 interface GigabitEthernet0/0/0/0.1920
  address-family ipv6 unicast
```

Explanation

```
show isis ipv6 topology
show isis topology

# ios-xr
show isis database verbose | utility egrep "NLPID|00-00"

# ios-xe
show isis database verbose | in NLPID|00-00
```

IPv6 Single to Multitopology

Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.ipv6.st.cfg

# IOS-XR
configure
load config:labconfigs/isis.ipv6.st.cfg
commit replace
y
```

All routers have **IPv6 link-local addresses** enabled on their interfaces, and a loopback IPv6 address of 2001:db8::<router num>/128.

Single-topology IPv6 has already been setup. **Migrate** to **Multi-topology** for IPv6 without breaking IPv4 or IPv6 reachability.

Explanation

In phase one, we **first** enable **wide-metrics everywhere**. This is because MT on IOS-XE requires wide metrics. We enable it using the transition method, in which both narrow and wide metrics are generated and received.

```
# IOS-XE
router isis
  Metric-style transition

# IOS-XR
Router isis 1
  address-family ipv4
    Metric-style transition
```

Now we can remove the transition style and use only wide-metrics.

```
# IOS-XE
router isis
  metric-stydde

# IOS-XR
Router isis 1
  address-family ipv4
    Metric-style wide
```

In phase two we enable multi-topology on all routers but with the **transition** keyword. This forces the router to continue generating the IPv6 reachability TLV which is used in single-topology.

```
# IOS-XE
router isis
  add ipv6
   multi-topology transition

# IOS-XR
router isis 1
  address-family ipv6
    no single-topology !!! no transition mode on IOS-XR
```

Note that we lost reachability on XRv neighbors with this step because we had to enable multi-topology only. There is no “transition” mode in which XRv continues generating normal ST IPv6 prefix reachability TLVs. Specifically, when making the change first on routers R1-R6, and then on XR1, XR1 loses its reachability to XR2 prefixes. But it still has reachability to all routers R1-R6. We then turn ST off on XR2, and reachability is restored.

Now that all routers are running MT, we can turn the transition mode off.

```
# IOS-XE
router isis
  add ipv6
   multi-topology
```

Summary

The **transition** keyword for multi-topology is only available on IOS-XE. When you use this keyword, the router continues generating IPv6 reachability TLVs (236) in addition to the MT IPv6 reachable prefixes TLV (237). This allows other routers that are still running in ST mode to continue to work.

Wide Metrics

Load Metric

```
# CSR
configure replace tftp://10.0.0.2/configs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace confirmed minutes 1
commit
```

Narrow metric TLVs is **6 bits**, so the max value is **63**.

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int eth0/1.12
R1(config-subif)#isis metric 100
Warning: for metrics greater than 63, 'metric-style wide'  should be configured on level-1-2, or it wil
l be capped at 63.
R1(config-subif)#
```

Redistribute a route on R1 into level 1 and notice the behavior.

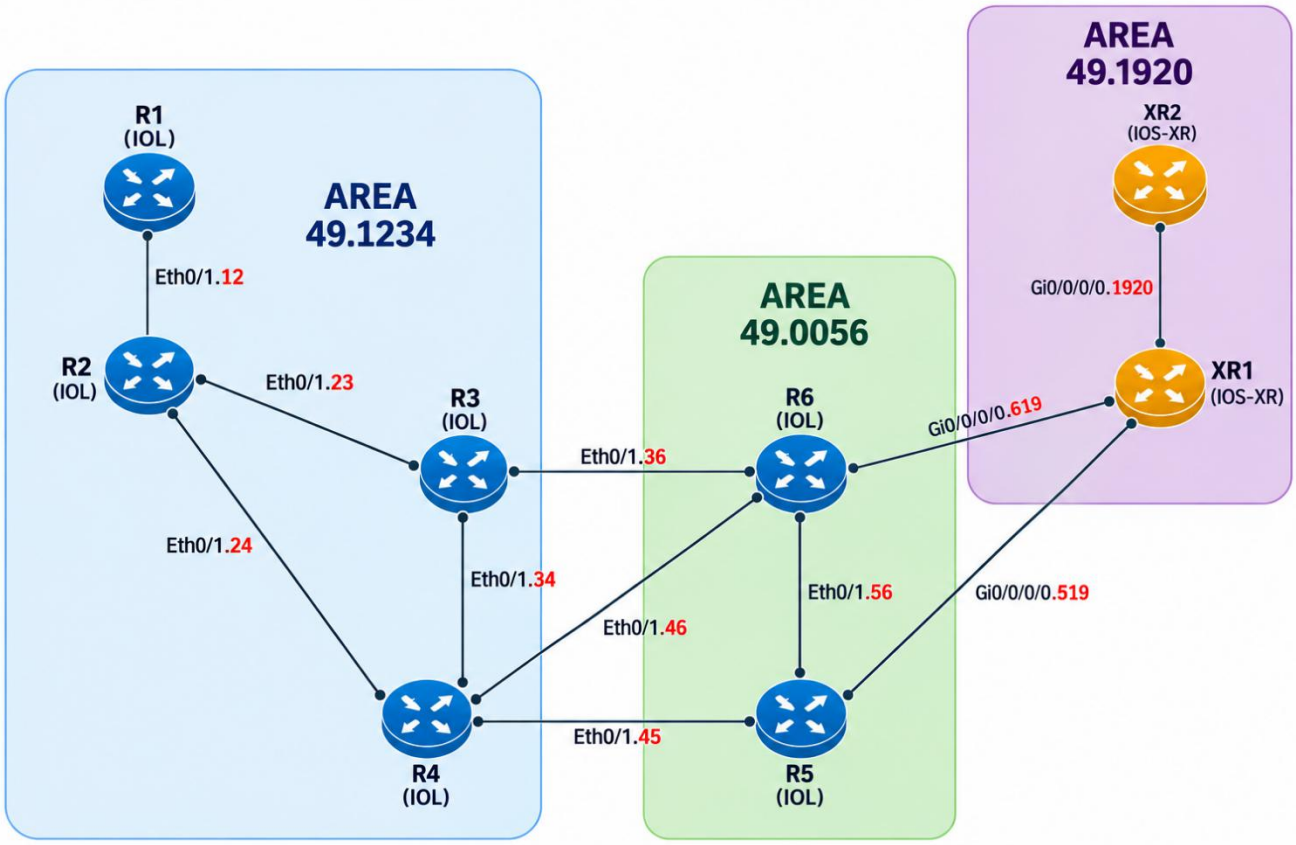
```
# R1
int lo100
 ip add 100.1.1.1 255.255.255.255
!
route-map LO100
 match int lo100
!
router isis
 redistribute connect route-map LO100 level-1
```

```
R1#show isis database R1.00-00 verbose


IS-IS Level-1 LSP R1.00-00
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x00000003   0xD956                1189/*         0/0/0
  Area Address: 49.1234
  NLPID:        0xCC
  Hostname: R1
  Metric: 10         IS R2.00
  IP Address:   1.1.1.1
  Metric: 0          IP 1.1.1.1 255.255.255.255
  Metric: 10         IP 10.1.2.0 255.255.255.0
  Metric: 0          IP-External 100.1.1.1 255.255.255.255
R1#
```

```
# R1
#R1
router isis
 metric-style transition
```

Route Filtering



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace
y
```

On **XR2**, configure **Lo20** with IP **200.20.20.20/32**. Redistribute this into ISIS with **tag=20**.

On **R3** and **R4**, filter routes with **tag=20** from being redistributed into **L1**. Allow all other L2 routes to be redistributed.

Answer

First we must switch to **wide-style metrics** to use **tags**. The narrow metric external IP reachability TLV does not have a tag field.

Redistribute with correct tag

```
# XR2

int Lo20
 ip add 200.20.20.20/32
!
route-policy lo20
 if destination in ( 200.20.20.20/32 ) then
  set tag 20
 endif
end-policy
!
router isis 1
 address-family ipv4 unicast
  redistribute connected level-1 route-policy lo20
```

On R3 and R4 we redistribute all L2 routes into L1 except for tag 20:

```
# R3, R4
route-map nolo20 deny 5
 match tag 20
route-map nolo20 permit 10
!
router isis
 redistribute isis ip level-2 into level-1 route-map nolo20
```

Backdoor Link

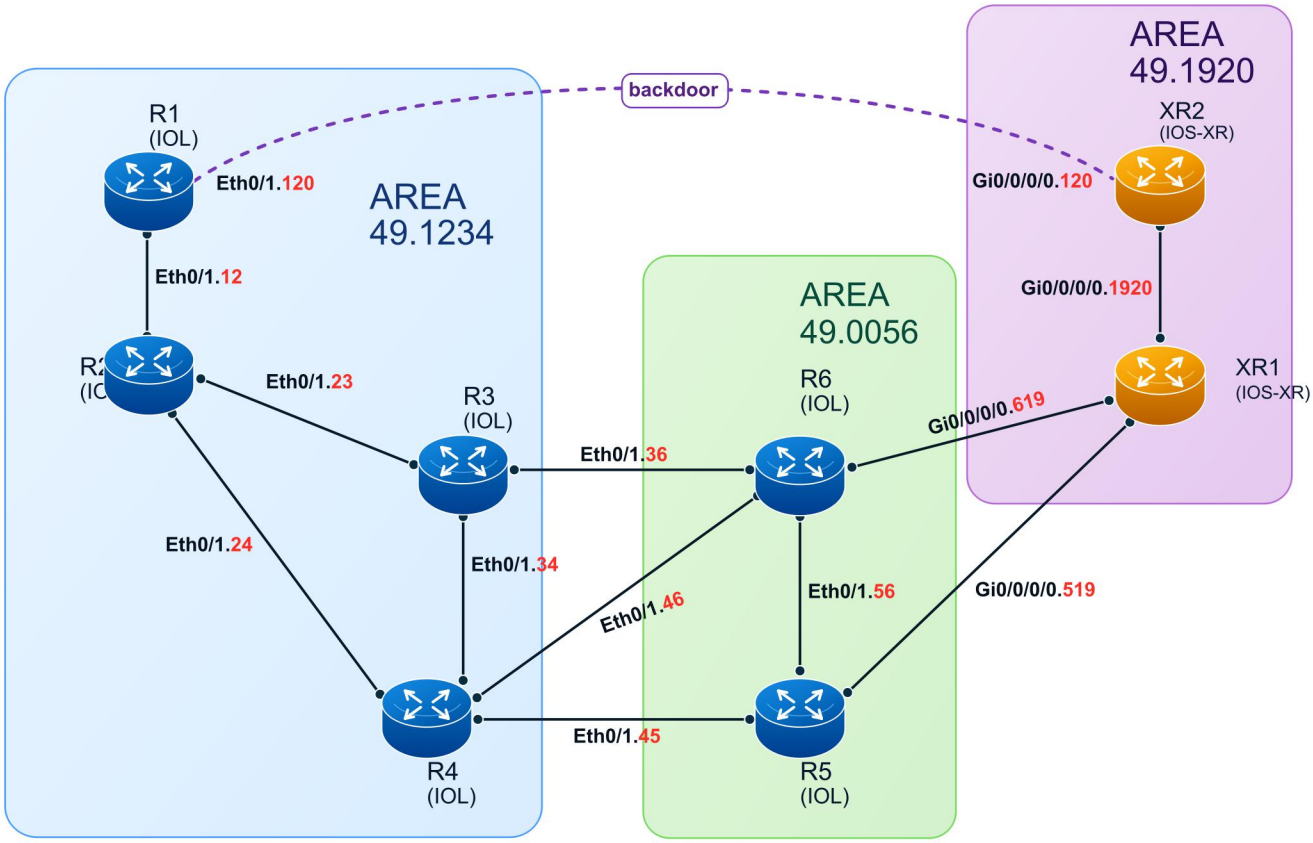


Figure 0-2 ISIS Backdoor

Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace
y
```

- Configure an additional link between **R1** and **XR2**, using **vlan 120** with subnet **20.1.20.0/24**.
- Configure this link so that area **49.1234** and **49.1920** can reach each other directly **without** traversing the **L2 backbone**.
- Do not let traffic destined for any other areas run over this link.

Answer

```
# R1
router isis
```

```
 is-type level-1-2 backdoor

# XR2
router isis 1
 address-family ipv4 unicast
  attached-bit send never-set
```

Explanation

The backdoor feature configures a L1/L2 router to do two things:

1. Not set the ATT bit in L1 LSPs
2. Continue installing a default route via other L1 routers that have the ATT bit set

On IOS-XE this is configured with **is-type level-1-2 backdoor**. On IOS-XR this is configured using **attached-bit send never-set**.

```
R1#show isis database

IS-IS Level-1 Link State Database:
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x00000004   0xE20B                1035/*         0/0/0
R2.00-00              0x00000002   0xAF5C                 369/1200      0/0/0
R3.00-00              0x00000004   0xCDAE                1158/1198      1/0/0
R4.00-00              0x00000004   0x3839                1181/1198      1/0/0
IS-IS Level-2 Link State Database:
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x00000004   0x094C                1077/*         0/0/0
XR2.00-00             0x00000003   0xCD85                1084/1191      0/0/0
XR2.05-00             0x00000001   0xFC9C                1084/1191      0/0/0

R1#show ip route isis | beg Gate
Gateway of last resort is 10.1.2.2 to network 0.0.0.0

i*L1  0.0.0.0/0 [115/20] via 10.1.2.2, 00:00:13, Ethernet0/1.12
      2.0.0.0/32 is subnetted, 1 subnets
i L1     2.2.2.2 [115/10] via 10.1.2.2, 00:10:15, Ethernet0/1.12
      3.0.0.0/32 is subnetted, 1 subnets
i L1     3.3.3.3 [115/20] via 10.1.2.2, 00:10:15, Ethernet0/1.12
      4.0.0.0/32 is subnetted, 1 subnets
i L1     4.4.4.4 [115/20] via 10.1.2.2, 00:10:15, Ethernet0/1.12
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
i L2     10.19.20.0/24 [115/20] via 10.1.20.20, 00:09:02, Ethernet0/1.120
      19.0.0.0/32 is subnetted, 1 subnets
i L2     19.19.19.19 [115/20] via 10.1.20.20, 00:09:02, Ethernet0/1.120
      20.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
i L1     20.2.3.0/24 [115/20] via 10.1.2.2, 00:10:15, Ethernet0/1.12
i L1     20.2.4.0/24 [115/20] via 10.1.2.2, 00:10:15, Ethernet0/1.12
i L2     20.20.20.20/32 [115/10] via 10.1.20.20, 00:09:02, Ethernet0/1.120

RP/0/RP0/CPU0:XR2#show isis database
Tue Oct 20 04:27:47.158 UTC

IS-IS 1 (Level-1) Link State Database
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd  ATT/P/OL
XR1.00-00             0x00000004   0xe811        1102 /1200         1/0/0
XR2.00-00           * 0x00000005   0xa69a        1079 /*            0/0/0
```

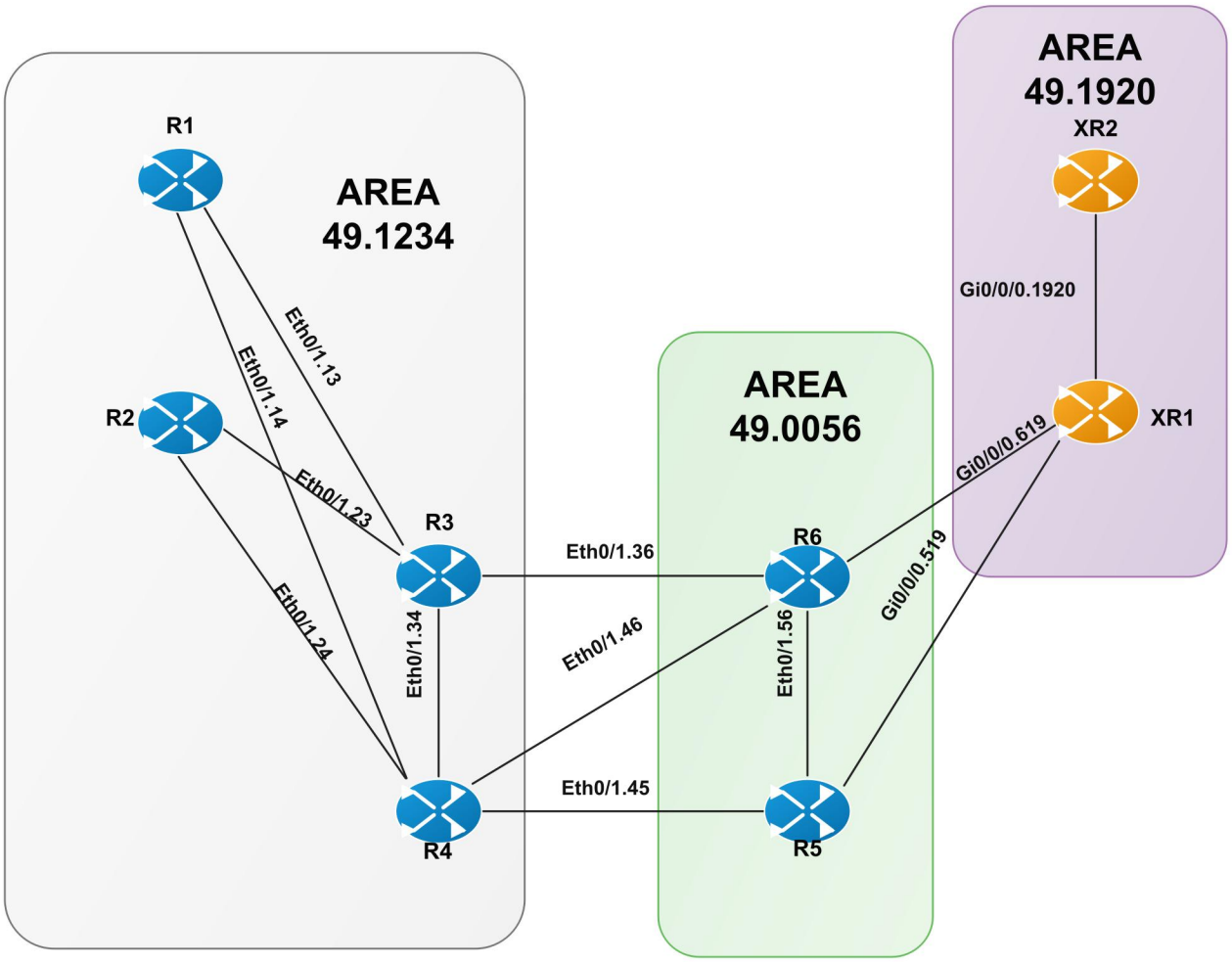
```
 Total Level-1 LSP count: 2     Local Level-1 LSP count: 1

IS-IS 1 (Level-2) Link State Database
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd  ATT/P/OL
R1.00-00              0x00000004   0x094c        989  /1200         0/0/0
XR2.00-00           * 0x00000003   0xcd85        989  /*            0/0/0
XR2.05-00             0x00000001   0xfc9c        989  /*            0/0/0

RP/0/RP0/CPU0:XR2#show route isis
Tue Oct 20 04:32:15.705 UTC

i*L1 0.0.0.0/0 [115/10] via 10.19.20.19, 00:06:28, GigabitEthernet0/0/0/0.1920
i L2 1.1.1.1/32 [115/10] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
i L2 2.2.2.2/32 [115/20] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
i L2 3.3.3.3/32 [115/30] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
i L2 4.4.4.4/32 [115/30] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
i L2 10.1.2.0/24 [115/20] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
i L1 19.19.19.19/32 [115/10] via 10.19.20.19, 00:18:39, GigabitEthernet0/0/0/0.1920
i L2 20.2.3.0/24 [115/30] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
i L2 20.2.4.0/24 [115/30] via 10.1.20.1, 00:07:58, GigabitEthernet0/0/0/0.120
```

Non-Optimal Intra-Area Routing



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.non-optimal.intra-area.cfg

# IOS-XR
configure
load config:labconfigs/isis.non-optimal.intra-area.cfg
commit replace
y
```

R1 is now connected to both **R3** and **R4**.

Shutdown the **R1-R3 link** and do a traceroute to **1.1.1.1** from **R3**. Traffic is from R3 to R1 is flowing to **R2-R4-R1** instead of **R4-R1**. Explain the issue and find a way to fix it so that traffic takes the path **R4-R1**.

Answer

```
# R3, R4
interface eth0/1.34
   no isis circuit-type
```

Explanation

```
# R3
interface eth0/1.13
  shutdown

R3#traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 20.2.3.2 0 msec 0 msec 0 msec
  2 20.2.4.4 0 msec 0 msec 0 msec
  3 20.1.4.1 0 msec 0 msec 0 msec

R3#show ip route 1.1.1.1
Routing entry for 1.1.1.1/32
  Known via "isis", distance 115, metric 30, type level-1
  Redistributing via isis
  Advertised by isis (self originated)
  Last update from 20.2.3.2 on Ethernet0/1.23, 00:00:10 ago
  Routing Descriptor Blocks:
  * 20.2.3.2, from 1.1.1.1, 00:00:10 ago, via Ethernet0/1.23
      Route metric is 30, traffic share count is 1

R3#show isis nei

Tag null:
System Id       Type Interface     IP Address      State Holdtime Circuit Id
R2              L1   Et0/1.23      20.2.3.2        UP    22       08
R4              L2   Et0/1.34      20.3.4.4        UP    25       07
R6              L2   Et0/1.36      20.3.6.6        UP    27       09

R3#show isis rib 1.1.1.1

IPv4 local RIB for IS-IS process

1.1.1.1/32
  [115/L1/30] via 20.2.3.2(Ethernet0/1.23), from 1.1.1.1, tag 0
              (installed)
  [115/L2/20] via 20.3.4.4(Ethernet0/1.34), from 4.4.4.4, tag 0


# R3
interface Ethernet0/1.34
 no isis circuit-type

R3#show isis neighbors

Tag null:
System Id       Type Interface     IP Address      State Holdtime Circuit Id
R2              L1   Et0/1.23      20.2.3.2        UP    29       08
R4              L1L2 Et0/1.34      20.3.4.4        UP    28       07
R6              L2   Et0/1.36      20.3.6.6        UP    23       09
```

```
R3#show isis rib 1.1.1.1

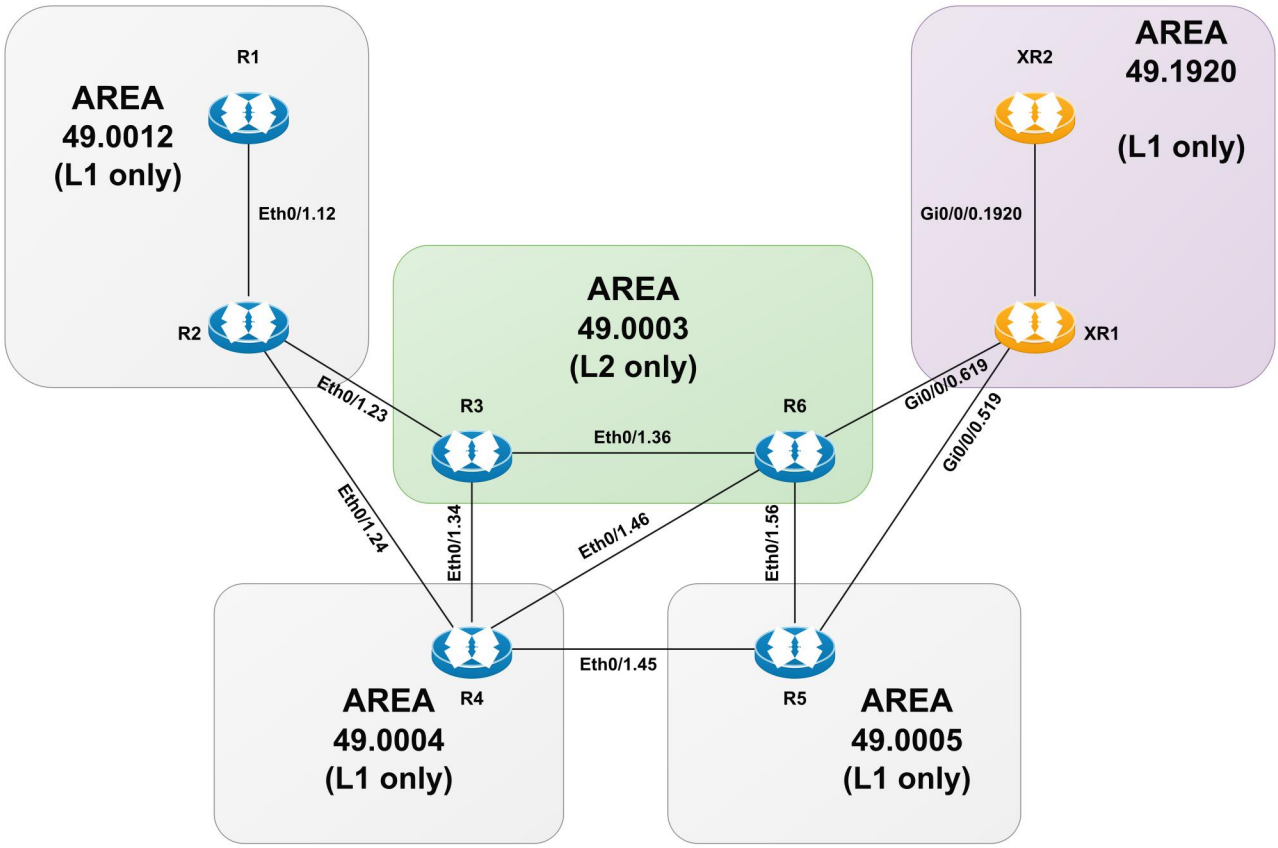
1.1.1.1/32
  [115/L1/20] via 20.3.4.4(Ethernet0/1.34), from 1.1.1.1, tag 0
              (installed)
  [115/L2/20] via 20.3.4.4(Ethernet0/1.34), from 4.4.4.4, tag 0
R3#
```

The reason for this **non-optimal** routing is because the **R3-R4** link is **level-2-only**.

Routers **prefer intra-area routes** over inter-area routes. This means an **L1 route is preferred** over an L2 route.

Even though R3 and R4 are in the same area, they only have an L2 adjacency. The route to 1.1.1.1 via R4 is an L2 route, which although a shorter metric, loses against the higher metric L1 route via R2.

Multi Area



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.multi.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.multi.area.cfg
commit replace
y
```

All routers are pre-configured in their respective ISIS areas. **All links** are **L1-only** except for **Eth0/1.36** on **R3** and **R6**.

Without changing the **L1-only areas** or changing the **circuit types**, fix the design so that **inter-area routing** is achieved.

Answer

```
# R3
router isis
 net 49.0012.0000.0000.0003.00
 net 49.0004.0000.0000.0003.00

# R6
router isis
 net 49.1920.0000.0000.0006.00
 net 49.0005.0000.0000.0006.00
```

Explanation

The solution to this involves setting **multiple NET addresses** on **R3** and **R6**. This allows these routers to participate in multiple ISIS areas.

The problem is that ISIS by default allows up to 3 areas on a single router. This is a negotiated value in the ISIS hello. R6 will need to participate in 4 areas. So first, we would change the maximum areas supported on all routers to 4.

Unfortunately, **IOS-XR** does **not support** a **different max-area-address number** besides 0 or 3. (0 implies a value of 3). So we cannot use this solution.

Instead we will have to **leave** the **default** at **3 max addresses** and **not create** an adjacency between **R4-R6**. We will configure R3 and R6 with additional NET addresses.

Authentication 1

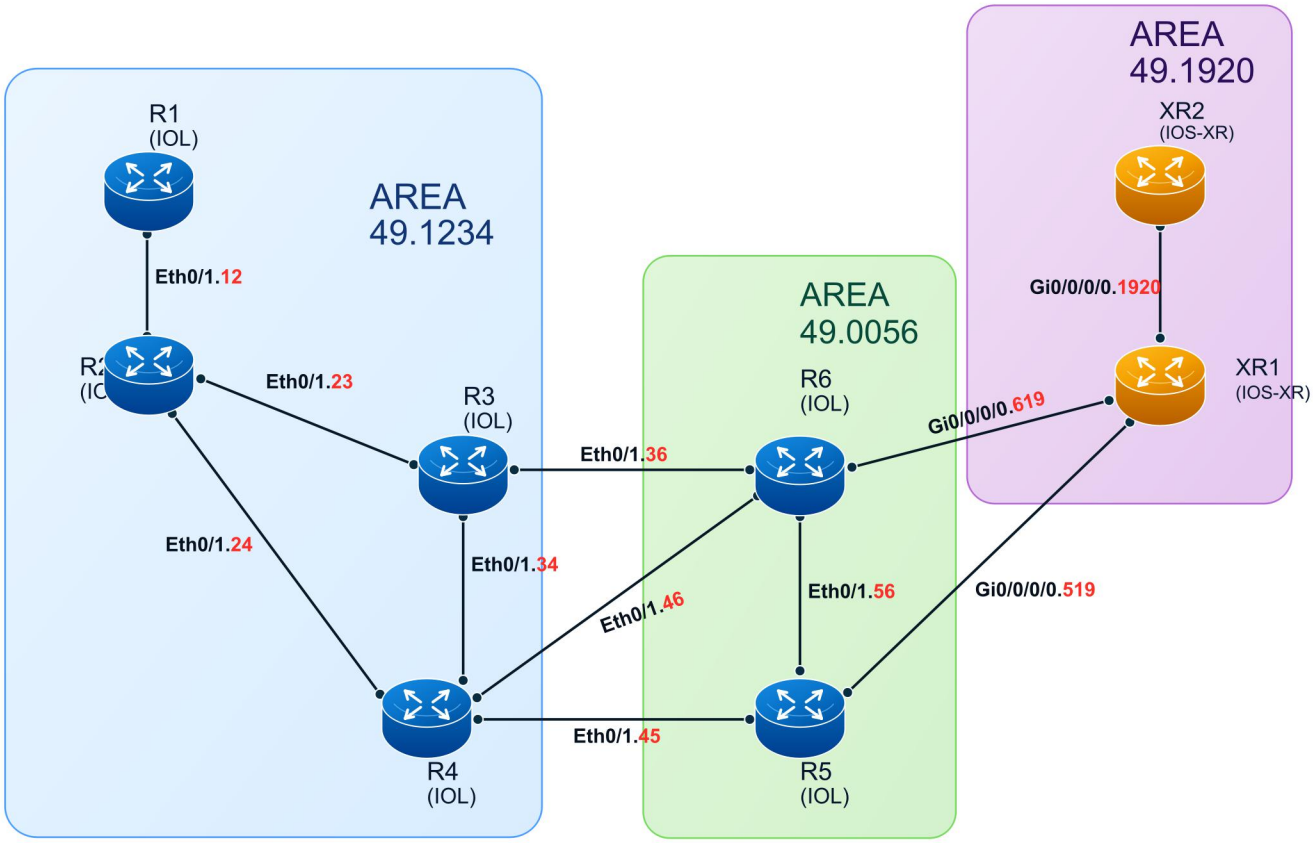


Figure 0-3 ISIS InterArea

Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace
y
```

Configure area **49.1234** as follows:

- **R1-R2** should use **HELLO123** as the hello password using the **legacy syntax**.
- **R2-R3-R4** should all use **HELLO234** as the password using **MD5**.
- The **area** should authenticate **L1 LSPs** and **SNPs** using "**AREA1234**" in **clear text**, but they should all **be accepted** if they do **not contain** authentication. Use the legacy syntax for this.

Answer

```
# R1
int Eth0/1.12
 isis password HELLO123
# R2
int Eth0/1.12
 isis password HELLO123
int Eth0/1.23
 isis authentication mode md5
 isis authentication key-chain ISIS_HELLO
int Eth0/1.24
 isis authentication mode md5
 isis authentication key-chain ISIS_HELLO
!
key chain ISIS_HELLO
 key 1
  key-string HELLO234

# R3
int Eth0/1.23
 isis authentication mode md5
 isis authentication key-chain ISIS_HELLO
int Eth0/1.34
 isis authentication mode md5
 isis authentication key-chain ISIS_HELLO
!
key chain ISIS_HELLO
 key 1
  key-string HELLO234
!
# R4
int Eth0/1.24
 isis authentication mode md5
 isis authentication key-chain ISIS_HELLO
int Eth0/1.34
 isis authentication mode md5
 isis authentication key-chain ISIS_HELLO
!
key chain ISIS_HELLO
 key 1
  key-string HELLO234
!

# R1, R2, R3, R4

router isis
 area-password AREA1234 authenticate snp send-only
```

```
# R1
int eth1.12
  isis password HELLO123

debug isis authentication information
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: No authentication TLV found in received packet
ISIS-AuthInfo: Insert auth for Hello
```

```
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: No authentication TLV found in received packet
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: No authentication TLV found in received packet
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: No authentication TLV found in received packet
%CLNS-4-AUTH_FAIL: ISIS: Serial IIH authentication failed
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: Packet has authentication type 1
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: Packet has authentication type 1
%CLNS-5-ADJCHANGE: ISIS: Adjacency to R2 (Ethernet0/1.12) Up, new adjacency

router isis
 area-password AREA1234 authenticate snp send-only

ISIS-AuthInfo: Packet has authentication type 1
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Insert auth for SNP or LSP
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: Packet has authentication type 1
ISIS-AuthInfo: Insert auth for Hello
ISIS-AuthInfo: Insert auth for SNP or LSP
ISIS-AuthInfo: Check authentication for Hello packet
ISIS-AuthInfo: Packet has authentication type 1
```

Authentication 2

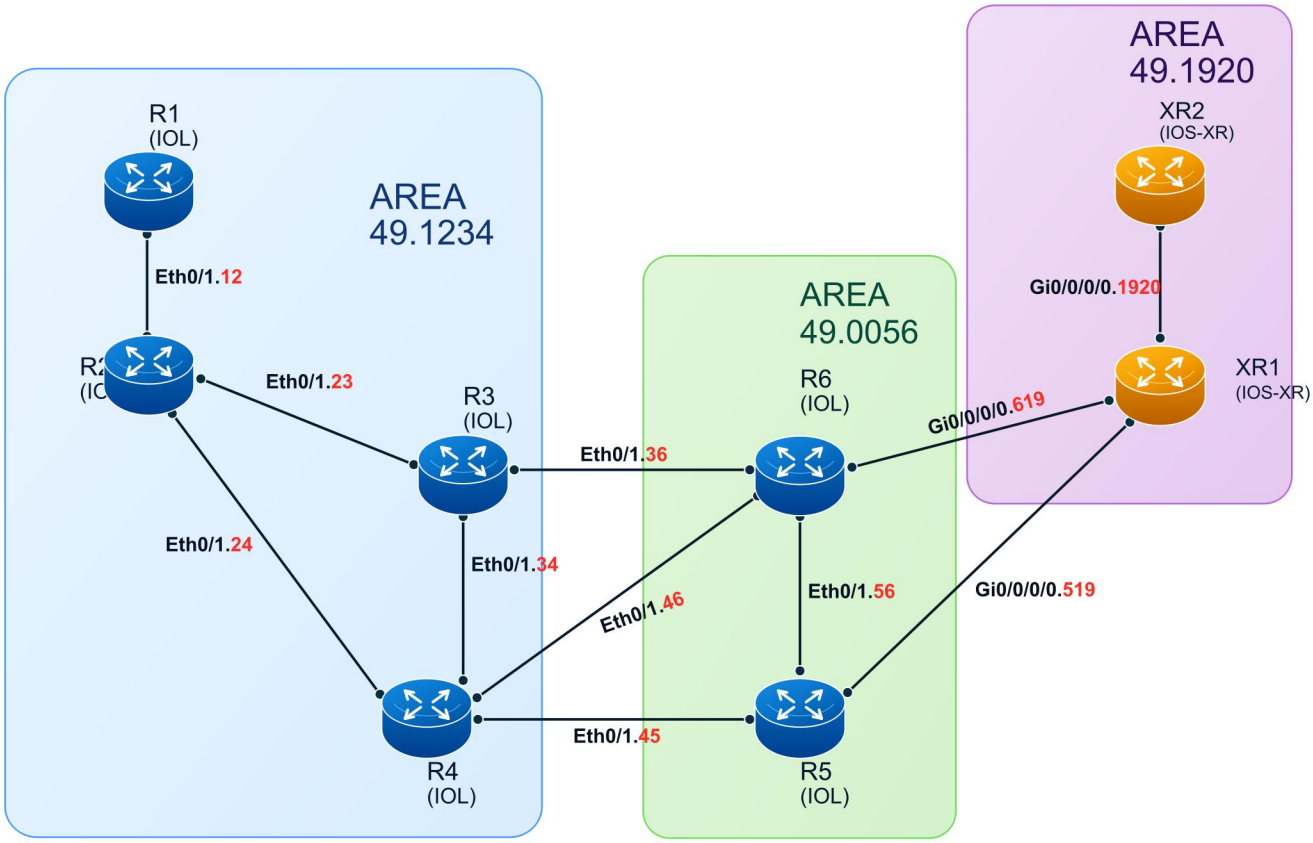


Figure 0-3 ISIS InterArea

Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace
y
```

Configure area **49.0056** as follows:

- **R5-R6** should use **HELLO56** as the hello password in **plain text** using the **new syntax**. Allow **received** Hellos to **not contain** authentication, or to contain wrong authentication.
- The **area** should authenticate **L1 LSPs** and **SNPs** using "**AREA56**" in clear **text**. LSPs and SNPs should be accepted **without authentication**. Use the new syntax for this.

Answer

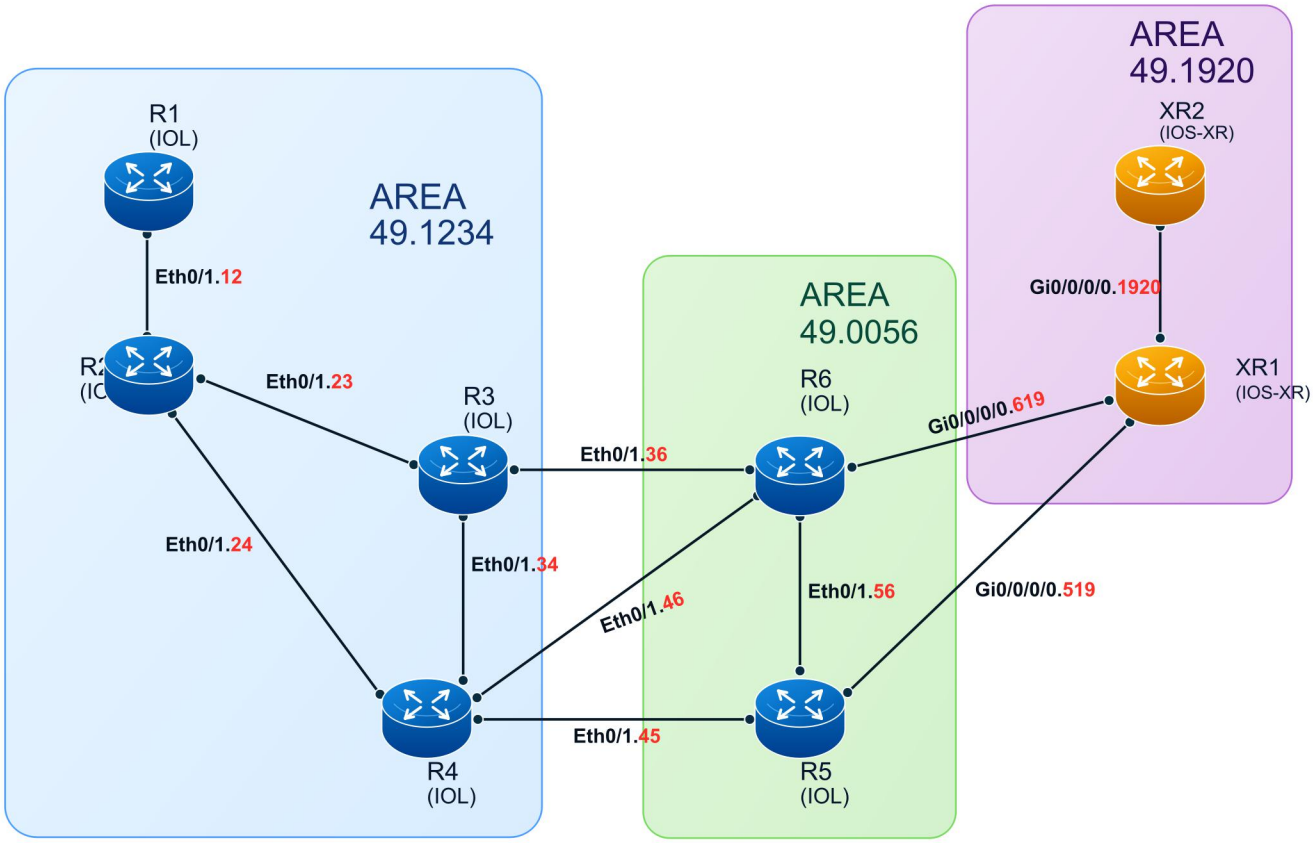
```
# R5, R6
key chain ISIS_HELLO
 key 1
  key-string HELLO56
!

int Eth0/1.56
 isis authentication mode text
 isis authentication key-chain ISIS_HELLO
 isis authentication send-only

key chain ISIS_AREA
 key 1
  key-string AREA56

router isis
 authentication mode text level-1
 authentication send-only level-1
 authentication key-chain ISIS_AREA level-1
```

Authentication 3



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace
y
```

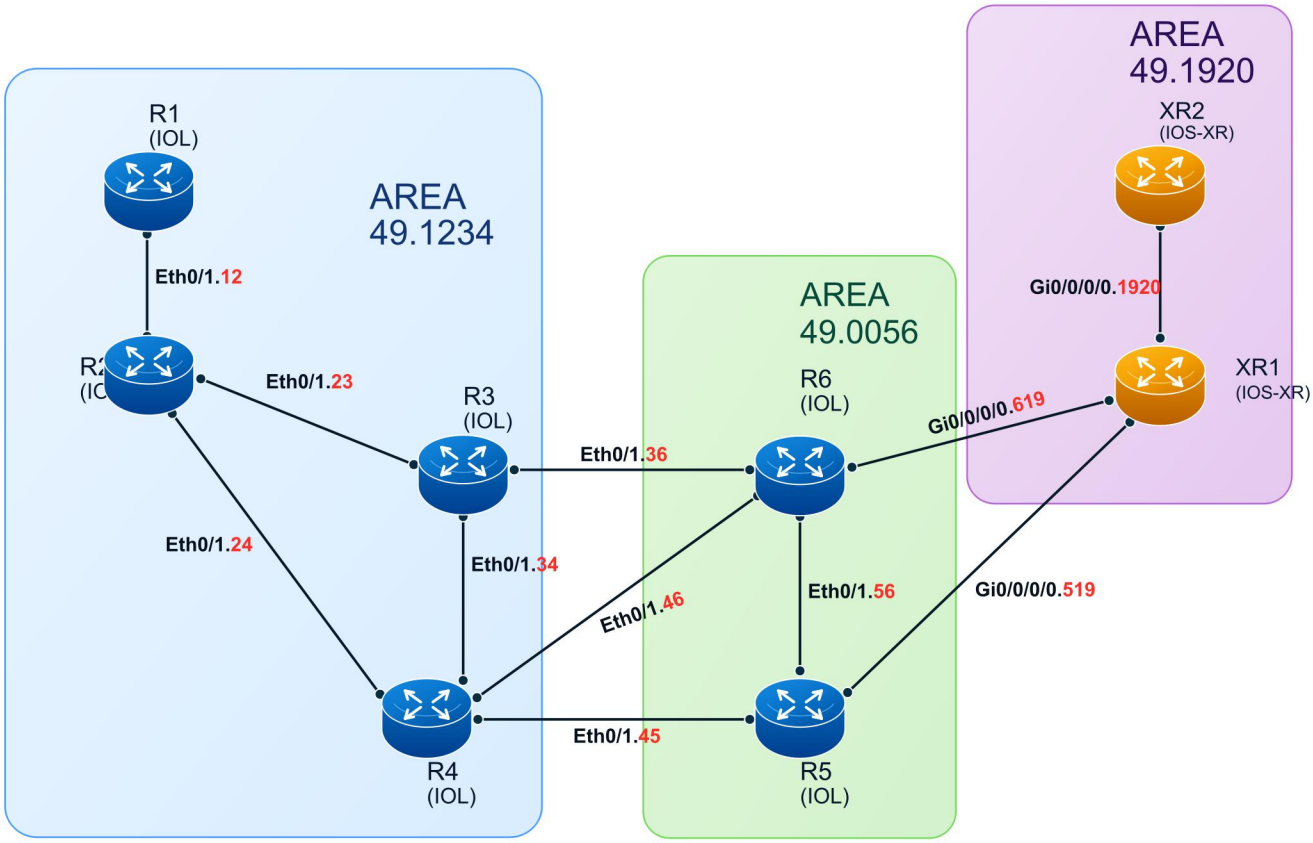
Configure area **49.1920** as follows:

- XR1-XR2 should use **HELLO1920** as the hello password using **MD5**.
- The **area** should authenticate **L1** LSPs and SNPs using "**AREA1920**" with **MD5**.

Answer

```
# XR1, XR2
router isis 1
 lsp-password hmac-md5 AREA1920 level 1
 !
 int gi0/0/0/0.1920
  hello-password hmac-md5 HELLO1920
```

Conditional ATT Bit



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.inter.area.cfg

# IOS-XR
configure
load config:labconfigs/isis.inter.area.cfg
commit replace
y
```

Configure **R4** so that it only sets the **attached bit** for area 49.1234 when the prefix **200.200.200.200/32** is present in the routing table.

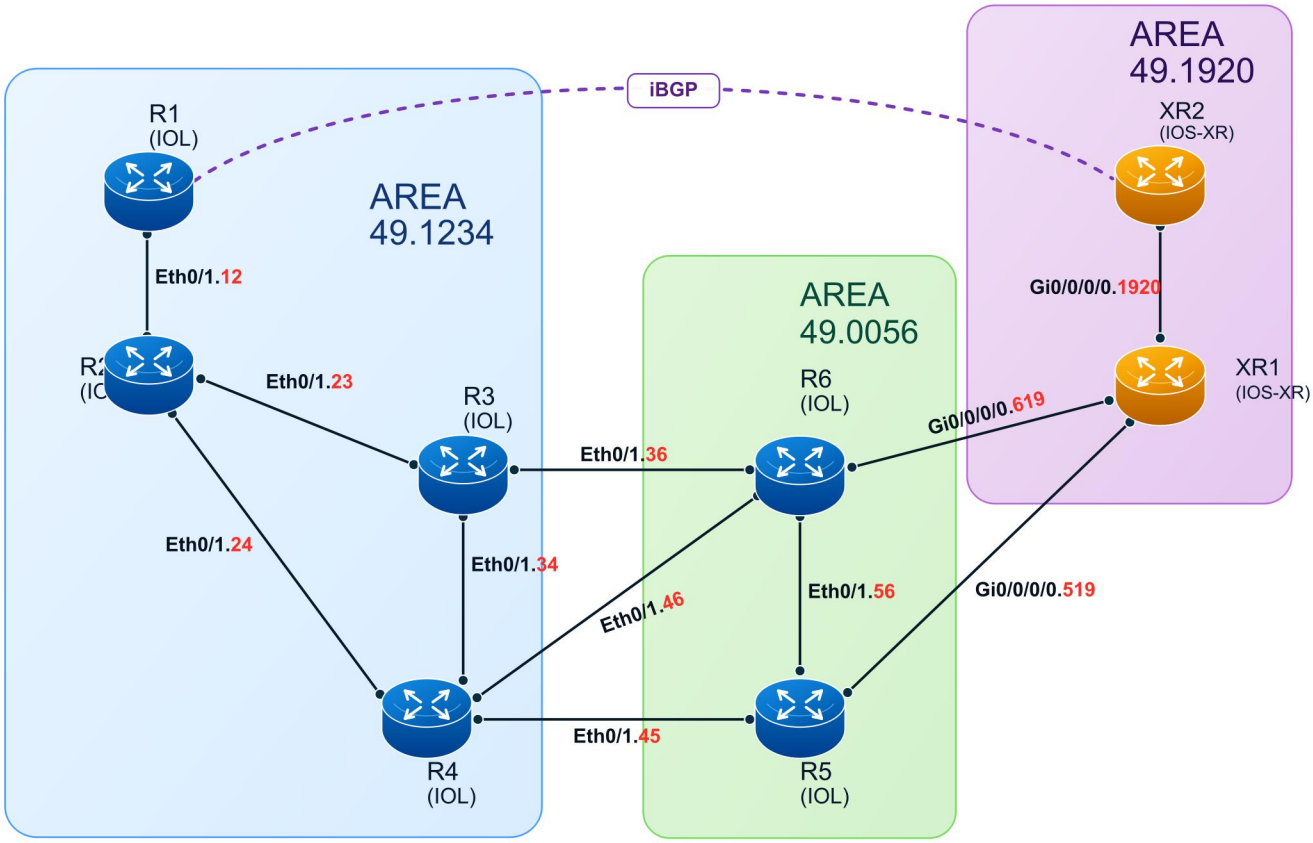
Answer

```
# R4

ip prefix-list lo200 seq 5 permit 200.200.200.200/32
route-map lo200 permit 10
 match ip address prefix-list lo200

router isis
 set-attached-bit route-map lo200
```

Troubleshoot iBGP



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.ibgp.tshoot.cfg force

# IOS-XR
configure
load config:labconfigs/isis.ibgp.tshoot.cfg
commit replace
```

An **iBGP** session is configured between **R1** and **XR2**. R1 is advertising a prefix (**100.100.100.100/32**). **LDP autoconfig** has been added so that traffic can be tunneled between R1 and XR2.

Answer

```
# R3, R4
router isis
 redistribute isis ip level-2 into level-1 route-map lo20

route-map lo20 permit 10
 match ip address prefix-list lo20
ip prefix-list lo20 seq 5 permit 20.20.20.20/32

# XR1
route-policy L2_TO_L1
```

```
 if destination in ( 1.1.1.1/32 ) then pass endif
end-policy
!
router isis 1
 add ipv4 uni
  propagate level 2 into level 1 route-policy L2_TO_L1
```

Explanation

IOS-XE and IOS-XR **cannot** use the **0/0 route** to establish a **BGP session** with a peer.

```
R1#ping 20.20.20.20 source 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.20.20.20, timeout is 2 seconds:
Packet sent with a source address of 1.1.1.1
!!!!!

R1#show ip route 20.20.20.20
% Subnet not in table


R1#show bgp ipv4 unicast neighbors 20.20.20.20 | in route
  BGP version 4, remote router ID 0.0.0.0
  Address tracking is enabled, the RIB does have a route to 20.20.20.20


RP/0/RP0/CPU0:XR2#show bgp ipv4 unicast neighbors 1.1.1.1 | in route
Sat Oct 24 20:44:05.058 UTC
 Remote router ID 0.0.0.0
  BGP state = Idle (No route to multi-hop neighbor)
  Advertise routes with local-label via Unicast SAFI

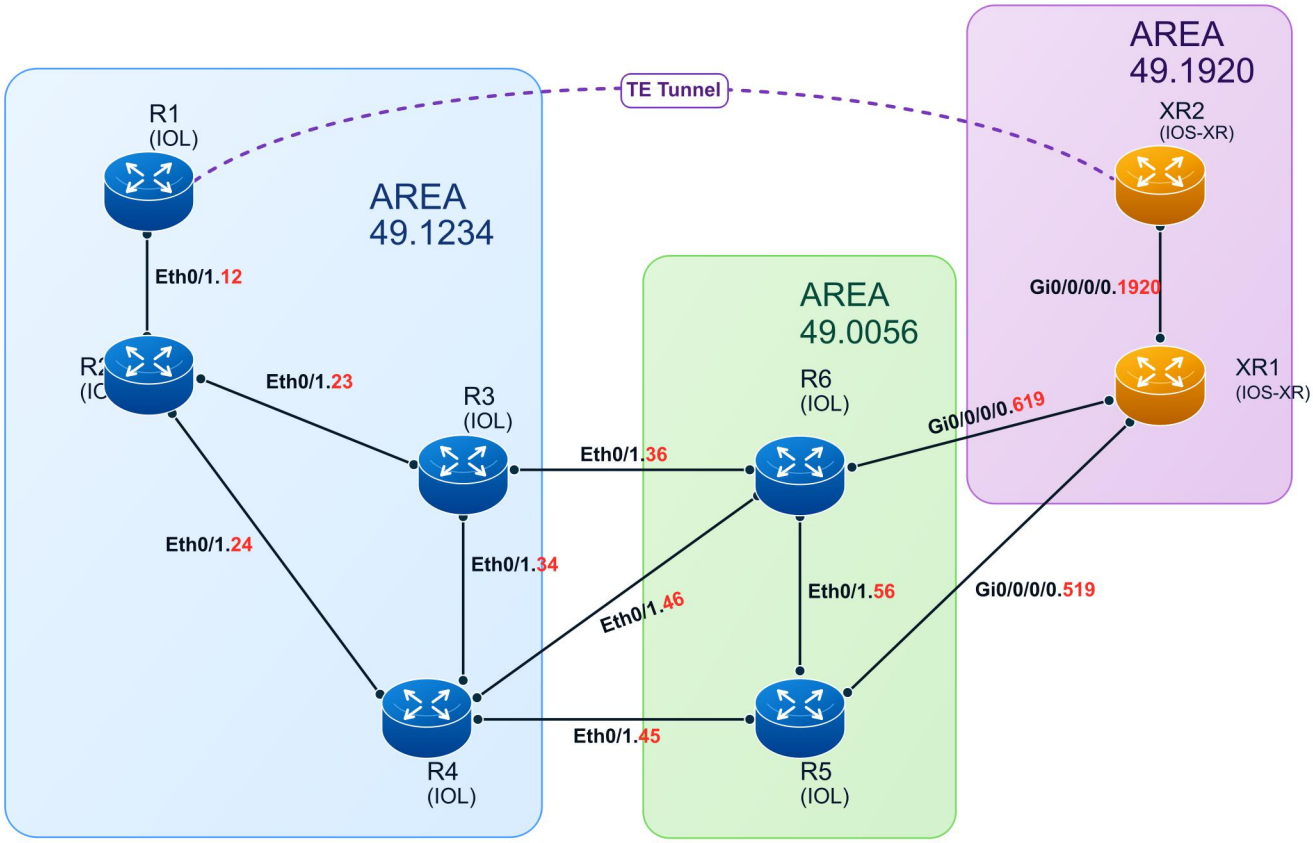
R1#debug ip bgp
BGP debugging is on for address family: IPv4 Unicast
R1#
BGP: 20.20.20.20 Active open failed - no route to peer, open active delayed 11264ms (35000ms
max, 60% jitter)
```

The bgp peer comes up after R3 and R4 **leaks 20.20.20.20/32** into **level-1**. But On **IOS-XR**, a **nexthop** cannot be resolved through the **0/0** route. So on **XR2**, we need to leak **1.1.1.1/32** into level-1.

```
RP/0/RP0/CPU0:XR2#
RP/0/RP0/CPU0:XR2#show bgp ipv4 unicast 100.100.100.100/32
Sat Oct 24 20:56:17.589 UTC
BGP routing table entry for 100.100.100.100/32
Versions:
  Process           bRIB/RIB   SendTblVer
  Speaker                 0            0
Last Modified: Oct 24 20:49:08.474 for 00:07:09
Paths: (1 available, no best path)
  Not advertised to any peer
  Path #1: Received by speaker 0
  Not advertised to any peer
  Local
    1.1.1.1 (inaccessible) from 1.1.1.1 (100.100.100.100)
      Origin IGP, metric 0, localpref 100, valid, internal
      Received Path ID 0, Local Path ID 0, version 0
```

This is **not the case** with **IOS-XE**. Prove this with injecting routes on XR2 and stop leaking 20.20.20.20/32 on R3 and R4.

Troubleshoot TE Tunnel



Load Config

```
# IOS-XE
config replace unix:labconfigs/isis.with.te.tshoot.init.cfg force

# IOS-XR
configure
load config:labconfigs/isis.with.te.tshoot.init.cfg
commit replace
```

A **TE tunnel** is setup on **R1** with destination **XR2**. The tunnel is not coming up. **Without making any** changes to any **other router** besides R1, bring the tunnel up.

Do the same for the **TE tunnel** on **XR2** with destination **R1**.

Answer

```
# R1
mpls traffic-eng path-selection overload allow middle

# XR2
mpls traffic-eng
 path-selection
  ignore overload mid
```

Explanation

R2 has the **overload** bit set.

```
R1#show mpls traffic-eng tunnels

P2P TUNNELS/LSPs:

Name: R1_t0                                (Tunnel0) Destination: 20.20.20.20
  Status:
    Admin: up         Oper: down   Path: not valid   Signalling: Down
    path option 1, type dynamic

  Config Parameters:
    Bandwidth: 0        kbps (Global)  Priority: 7  7   Affinity: 0x0/0xFFFF
    Metric Type: TE (default)
    Path-selection Tiebreaker:
      Global: not set   Tunnel Specific: not set   Effective: min-fill (default)
    Hop Limit: disabled
    Cost Limit: disabled
    Path-invalidation timeout: 10000 msec (default), Action: Tear
    AutoRoute: disabled LockDown: disabled Loadshare: 0 [0] bw-based
    auto-bw: disabled
    Fault-OAM: disabled, Wrap-Protection: disabled, Wrap-Capable: No

  History:
    Tunnel:
      Time since created: 6 minutes, 49 seconds
      Time since path change: 19 seconds
      Number of LSP IDs (Tun_Instances) used: 21
    Prior LSP: [ID: 19]
      ID: path option 1 [21]
      Removal Trigger: path verification failed
      Last Error: CTRL:: No path to destination, 0000.0000.0020.00 (no path)

P2MP TUNNELS:

P2MP SUB-LSPS:

R1#show mpls traffic-eng topology brief
My_System_id: 0000.0000.0001.00 (isis  level-2)

Signalling error holddown: 10 sec Global Link Generation 19

IGP Id: 0000.0000.0001.00, MPLS TE Id:1.1.1.1 Router Node  (isis  level-2)
      link[0]: Broadcast, DR: 0000.0000.0002.01, nbr_node_id:2, gen:12
      frag_id: 0, Intf Address: 10.1.2.1
      TE metric: 10, IGP metric: 10, attribute flags: 0x0
      SRLGs: None

IGP Id: 0000.0000.0002.00, MPLS TE Id:2.2.2.2 Router Node (NO SPF) (isis  level-2)
      link[0]: Broadcast, DR: 0000.0000.0003.01, nbr_node_id:4, gen:13
      frag_id: 0, Intf Address: 20.2.3.2
      TE metric: 10, IGP metric: 10, attribute flags: 0x0
      SRLGs: None

      link[1]: Broadcast, DR: 0000.0000.0002.01, nbr_node_id:2, gen:13
      frag_id: 0, Intf Address: 10.1.2.2
      TE metric: 10, IGP metric: 10, attribute flags: 0x0
```

```
      SRLGs: None

      link[2]: Broadcast, DR: 0000.0000.0004.01, nbr_node_id:5, gen:13
      frag_id: 0, Intf Address: 20.2.4.2
      TE metric: 10, IGP metric: 10, attribute flags: 0x0
      SRLGs: None


R1# show isis database

IS-IS Level-2 Link State Database:
LSPID                 LSP Seq Num  LSP Checksum  LSP Holdtime/Rcvd      ATT/P/OL
R1.00-00            * 0x00000019   0x08B8                593/*         0/0/0
R2.00-00              0x0000001A   0x9872                606/1199      0/0/1
R2.01-00              0x00000013   0x5A5D               1126/1199      0/0/0


R1(config)#mpls traffic-eng path-selection overload allow middle


R1#show mpls traffic-eng tunnels

P2P TUNNELS/LSPs:

Name: R1_t0                               (Tunnel0) Destination: 20.20.20.20
  Status:
    Admin: up         Oper: up     Path: valid       Signalling: connected
    path option 1, type dynamic (Basis for Setup, path weight 50)

  Config Parameters:
    Bandwidth: 0        kbps (Global)  Priority: 7  7   Affinity: 0x0/0xFFFF
    Metric Type: TE (default)
    Path-selection Tiebreaker:
      Global: not set   Tunnel Specific: not set   Effective: min-fill (default
```

IOS-XR gives us much better indication of the issue. We know that **some transit** node has the **OL bit set**, but not which node.

```
# XR2
RP/0/RP0/CPU0:XR2#show mpls traffic-eng tunnels role head
Sun Oct 25 22:50:08.364 UTC



Name: tunnel-te0  Destination: 1.1.1.1  Ifhandle:0x1c
  Signalled-Name: XR2_t0
  Status:
    Admin:    up Oper: down   Path: not valid   Signalling: Down

    path option 1,  type dynamic
    Last PCALC Error: Sun Oct 25 22:28:16 2026
      Info: No path to destination, 1.1.1.1 (overloaded transit node)
    G-PID: 0x0800 (derived from egress interface properties)
    Bandwidth Requested: 0 kbps  CT0
    Creation Time: Sun Oct 25 22:27:44 2026 (00:22:24 ago)
  Config Parameters:
    Source: router ID (default)
    Bandwidth:        0 kbps (CT0) Priority:  7  7 Affinity: 0x0/0xffff
    Metric Type: TE (global)
```

```
    Path Selection:
      Tiebreaker: Min-fill (default)
    Hop-limit: disabled
    Cost-limit: disabled
    Delay-limit: disabled
    Delay-measurement: disabled
    Path-invalidation timeout: 10000 msec (default), Action: Tear (default)
    AutoRoute: disabled  LockDown: disabled   Policy class: not set
    Forward class: 0 (not enabled)
    Forwarding-Adjacency: disabled
    Autoroute Destinations: 0

--------------------cut--------------------------------------------------

RP/0/RP0/CPU0:XR2#show mpls traffic-eng topology brief
Sun Oct 25 22:51:39.357 UTC
My_System_id: 0000.0000.0020.00 (IS-IS 1 level-2)
My_BC_Model_Type: RDM

Signalling error holddown: 10 sec Global Link Generation 1137

IGP Id: 0000.0000.0001.00, MPLS TE Id: 1.1.1.1 Router Node  (IS-IS 1 level-2)
 Interface Addresses:  1.1.1.1

  Link[0]:Broadcast, DR:0000.0000.0002.01, Nbr Node Id:22, gen:1094
      Frag Id:0, Intf Address:10.1.2.1, Intf Id:0
      Nbr Intf Address:0.0.0.0, Nbr Intf Id:0
      TE Metric:10, IGP Metric:10
      Delay metrics (uSec): Not present
      Loss metrics: Not present
      Bandwidth metrics (kbps): Not present
      Attribute Flags: 0x0
      Ext Admin Group:
          Length: 256 bits
          Value : 0x::
      Attribute Names:
      Switching Capability:None, Encoding:unassigned
      BC Model ID:RDM
      Physical BW:10000 (kbps), Max Reservable BW Global:0 (kbps)
      Max Reservable BW Sub:0 (kbps)

IGP Id: 0000.0000.0002.00, MPLS TE Id: 2.2.2.2 Router Node (Overloaded) (IS-IS 1
level-2)
 Interface Addresses:  2.2.2.2

  Link[0]:Broadcast, DR:0000.0000.0003.01, Nbr Node Id:24, gen:1095
      Frag Id:0, Intf Address:20.2.3.2, Intf Id:0


XR2
mpls traffic-eng
 path-selection
  ignore overload mid
 !
!


RP/0/RP0/CPU0:XR2#show mpls traffic-eng tunnels role head
```

```
Sun Oct 25 22:53:51.854 UTC

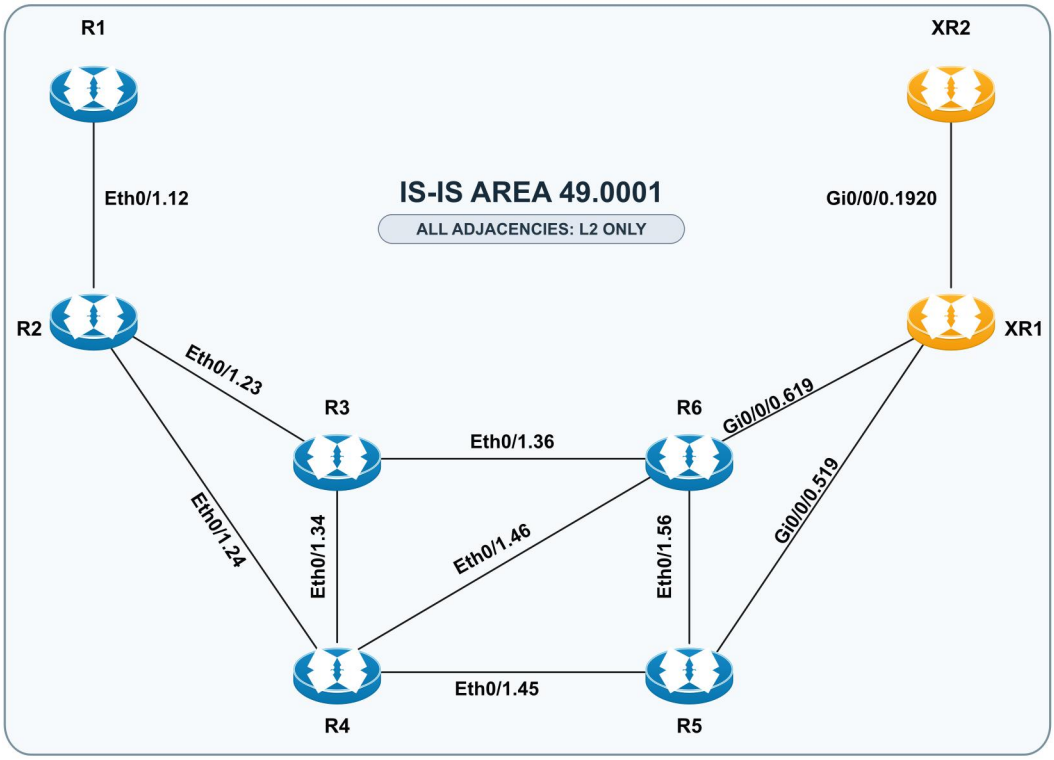


Name: tunnel-te0  Destination: 1.1.1.1  Ifhandle:0x1c
  Signalled-Name: XR2_t0
  Status:
    Admin:    up Oper:   up   Path:  valid   Signalling: connected

    path option 1,  type dynamic  (Basis for Setup, path weight 50)
      Accumulative metrics: TE 50 IGP 50 Delay 1500000
      Accumulative biased metrics: TE 50 IGP 50 Delay 1500000
    G-PID: 0x0800 (derived from egress interface properties)
    Bandwidth Requested: 0 kbps  CT0
    Creation Time: Sun Oct 25 22:27:44 2026 (00:26:07 ago)
  Config Parameters:
    Source: router ID (default)
    Bandwidth:        0 kbps (CT0) Priority:  7  7 Affinity: 0x0/0xffff
```

LDP

ECMP



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
 load config:labconfigs/isis.cfg
 commit replace confirmed minutes 1
 commit
```

ISIS is already pre-configured. Configure LDP on all routers using LDP IGP autoconfig.

Notice that R2 has ECMP routes in its RIB. How will R2 treat traffic sourced from 1.1.1.1 destined for 19.19.19.19 by default?

Answer

```
# IOL
router isis
  mpls ldp autoconfig

# IOS-XR
```

```
mpls ldp
router isis 1
 address-family ipv4 unicast
  mpls ldp auto-config
 !
commit
```

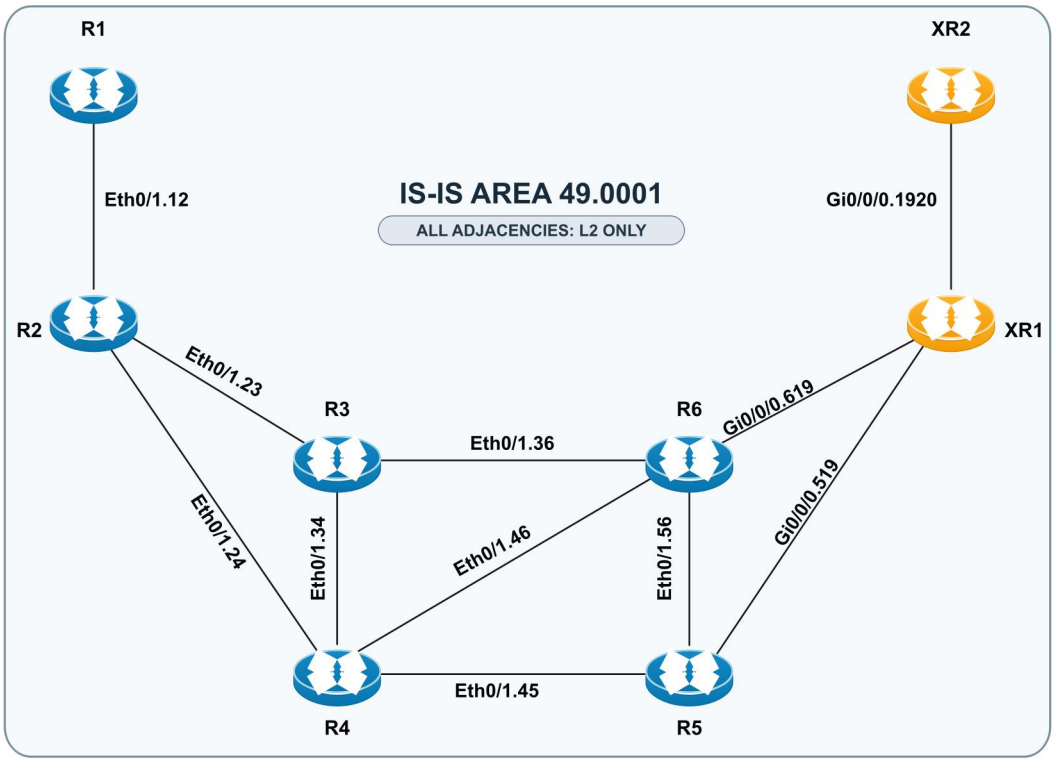
Explanation

```
# IOL
R2#show ip route 19.19.19.19
Routing entry for 19.19.19.19/32
  Known via "isis", distance 115, metric 30, type level-2
  Redistributing via isis
  Last update from 20.2.4.4 on Ethernet0/1.24, 00:18:57 ago
  Routing Descriptor Blocks:
    20.2.4.4, from 19.19.19.19, 00:18:57 ago, via Ethernet0/1.24
      Route metric is 30, traffic share count is 1
  * 20.2.3.3, from 19.19.19.19, 00:18:57 ago, via Ethernet0/1.23
      Route metric is 30, traffic share count is 1

R2#show ip cef 19.19.19.19 detail
19.19.19.19/32, epoch 0, per-destination sharing
  dflt local label info: global/22 [0x3]
  nexthop 20.2.3.3 Ethernet0/1.23 label 23-(local:22)
  nexthop 20.2.4.4 Ethernet0/1.24 label 23-(local:22)

R2#show mpls forwarding-table 19.19.19.19 detail
Local      Outgoing   Prefix           Bytes Label   Outgoing   Next Hop
Label      Label      or Tunnel Id     Switched      interface
22         23         19.19.19.19/32   0             Et0/1.23   20.2.3.3
       MAC/Encaps=18/22, MRU=1500, Label Stack{23}
       AABBCC000310AABBCC000210810000178847 00017000
       No output feature configured
    Per-destination load-sharing, slots: 0
           23         19.19.19.19/32   0             Et0/1.24   20.2.4.4
       MAC/Encaps=18/22, MRU=1500, Label Stack{23}
       AABBCC000410AABBCC000210810000188847 00017000
       No output feature configured
    Per-destination load-sharing, slots: 1
```

Static Routes



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

- Configure **LDP** on all routers using LDP IGP **autoconfig**.
- On **R1**, Configure a **static route** for **20.20.20.21/32** via **R2.**
- On **R2**, Configure a **static route** for **20.20.20.21/32** via **R3**.
- Do both routers advertise a label mapping for this prefix? If so, do they install an entry in the LFIB?

Answer

```
#R1-R6
router isis
 mpls ldp autoconfig

#R1
ip route 20.20.20.21 255.255.255.255 10.1.2.2

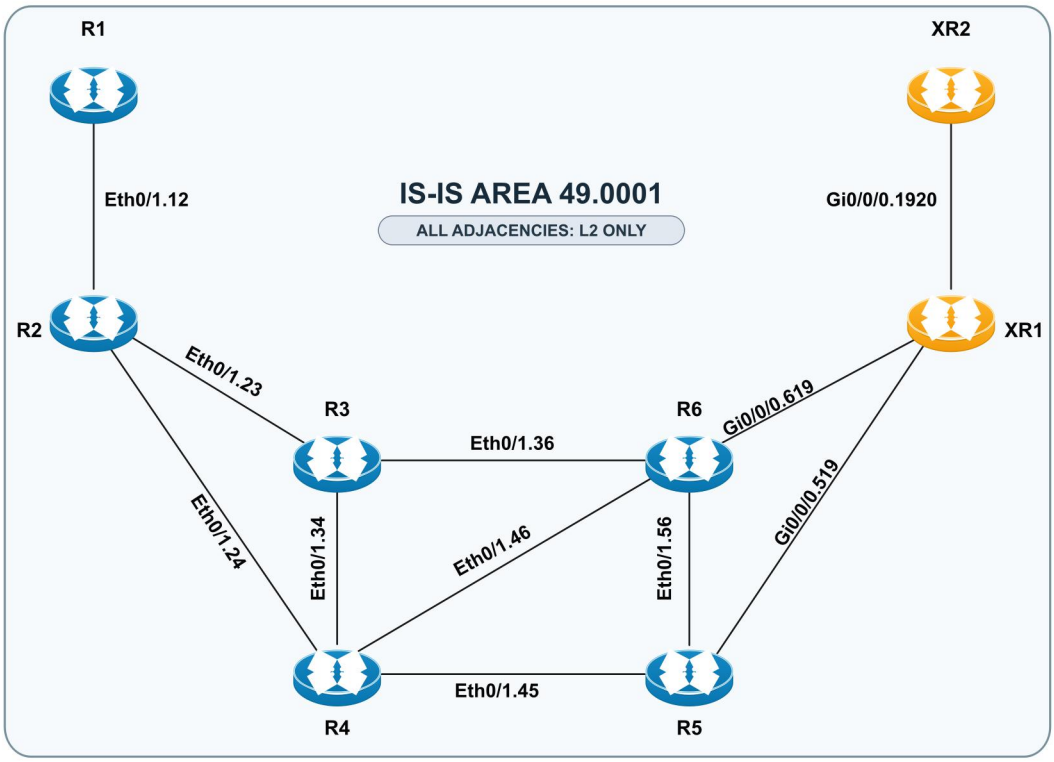
#R2
ip route 20.20.20.21 255.255.255.255 20.2.3.3
```

Static routes "**count**" as **IGP** routes to **LDP**. So labels are **locally allocated** for static routes, advertised as label bindings to all peers, and entries are installed in the forwarding table.

```
R1#show mpls ldp bindings 20.20.20.21 32
  lib entry: 20.20.20.21/32, rev 40
       local binding:  label: 33
       remote binding: lsr: 2.2.2.2:0, label: 31
R1#show mpls forwarding-table 20.20.20.21
Local      Outgoing   Prefix           Bytes Label   Outgoing   Next Hop
Label      Label      or Tunnel Id     Switched      interface
33         31         20.20.20.21/32   0             Et0/1.12   10.1.2.2

R2#show mpls ldp bindings 20.20.20.21 32
  lib entry: 20.20.20.21/32, rev 40
       local binding:  label: 31
       remote binding: lsr: 1.1.1.1:0, label: 33
R2#show mpls forwarding-table 20.20.20.21
Local      Outgoing   Prefix           Bytes Label   Outgoing   Next Hop
Label      Label      or Tunnel Id     Switched      interface
31         No Label   20.20.20.21/32   0             Et0/1.23   20.2.3.3
```

Timers



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

- Configure **LDP** on all routers using LDP IGP **autoconfig**.
- Configure **R1** and **R2** to use a **hello/hold** time of **1/3 seconds** for the **UDP discovery**.
- The **R2-R3** and the **R2-R4** neighborship should continue using **5/15 seconds**.
- Configure **XR1** and **R6** to detect **failure** for the **TCP session** after **20 seconds**. Configure only XR1 to achieve this.

Answer

```
# R1
mpls ldp discovery hello interval 1
mpls ldp discovery hello holdtime 3
# XR1
mpls ldp
 session holdtime 20
```

Explanation

The discovery hello/hold time is **only** configurable **globally**. It is not configurable per-interface or per-peer. However, we can use the fact that **LDP timers are negotiated** to our advantage.

```
# R1
mpls ldp discovery hello interval 1
mpls ldp discovery hello holdtime 3

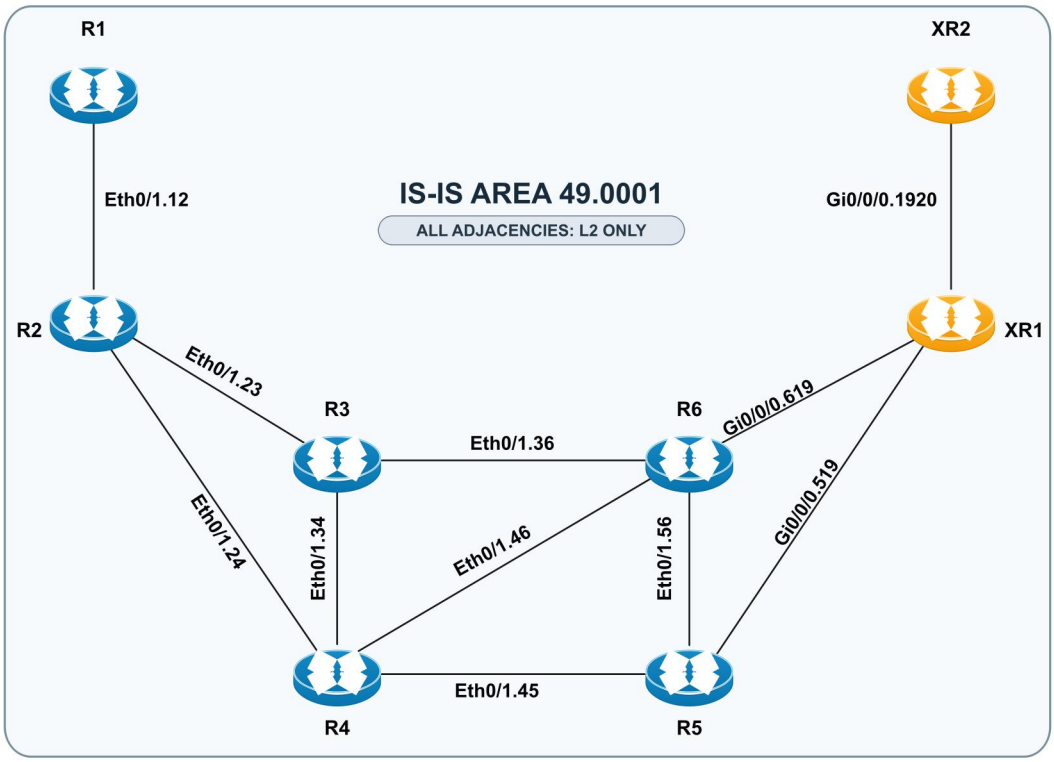
R2#show mpls ldp neighbor detail | in ms|Peer
    Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 2.2.2.2:0
      Up time: 00:05:00; UID: 1; Peer Id 0
          holdtime: 3000 ms, hello interval: 1000 ms
      Peer holdtime: 180000 ms; KA interval: 60000 ms; Peer state: estab
    Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 2.2.2.2:0
      Up time: 00:05:00; UID: 2; Peer Id 1
          holdtime: 15000 ms, hello interval: 5000 ms
      Peer holdtime: 180000 ms; KA interval: 60000 ms; Peer state: estab
    Peer LDP Ident: 4.4.4.4:0; Local LDP Ident 2.2.2.2:0
      Up time: 00:05:00; UID: 3; Peer Id 2
          holdtime: 15000 ms, hello interval: 5000 ms
      Peer holdtime: 180000 ms; KA interval: 60000 ms; Peer state: estab
```

Likewise, the LDP **TCP session timers** are negotiated to the **lowest value**. The **keepalive** is always **1/3 of the holdtime**. You cannot actually configure the session keepalive interval, only the session holdtime.

```
R6#show mpls ldp neighbor 19.19.19.19 detail | in holdtime
          holdtime: 15000 ms, hello interval: 5000 ms
      Peer holdtime: 20000 ms; KA interval: 6666 ms; Peer state: estab

R6#show mpls ldp neighbor 5.5.5.5 detail | in holdtime
          holdtime: 15000 ms, hello interval: 5000 ms
      Peer holdtime: 180000 ms; KA interval: 60000 ms; Peer state: estab
```

Authentication



Load Config

```
# IOL
configure replace unix:labconfigs/isis.cfg force

# IOS-XR
configure
load config:labconfigs/isis.cfg
commit replace confirmed minutes 1
commit
```

- Configure LDP on all routers using LDP IGP autoconfig.
- Configure **R1** and **R2** to use LDP authentication with a password of "**LDP12**"
- Configure **R3** and **R4** to use LDP authentication with a keychain called "**KC34**" and password of "**LDP34**"
- Configure **XR1** and **R6** to use an LDP password of "**LPD619**"
- Configure all other LDP sessions to use a password of "**LDPDEFAULT**"

Answer

```
# R1 <-> R2
mpls ldp nei 1.1.1.1 password LDP12
mpls ldp nei 2.2.2.2 password LDP12

# R3 <-> R4
```

```
key chain KC34
 key 1
  key-string LDP34

access-list 4 permit 4.4.4.4
mpls ldp password option 1 for 4 key-chain KC34

# R6 <-> XR1
mpls ldp nei 19.19.19.19 password LDP619

mpls ldp
 neighbor
  6.6.6.6:0 password clear LDP619


# IOS-XE
mpls ldp password fallback LDPDEFAULT

# IOS-XR
mpls ldp
 neighbor
  password clear LDPDEFAULT
```

Explanation

On **IOS-XR**, the change is **disruptive**. The session is torn down and re-established using the new TCP MD5 auth password.

```
RP/0/RP0/CPU0:XR1#show mpls ldp neighbor | in MD5
Wed Oct 28 09:46:20.171 UTC
  TCP connection: 20.20.20.20:51660 - 19.19.19.19:646; MD5 on
  TCP connection: 6.6.6.6:646 - 19.19.19.19:17018; MD5 on
  TCP connection: 5.5.5.5:646 - 19.19.19.19:29666; MD5 on

R6#show mpls ldp neighbor 19.19.19.19 detail | in Pass
       Password: not required, neighbor, in use
```

By default, LDP sessions are ***not*** **torn down** on **IOS-XE** when you make this change.

```
# R1 <-> R2

R1#show mpls ldp neighbor 2.2.2.2 password
    Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0
       TCP connection: 2.2.2.2.53629 - 1.1.1.1.646
       Password: not required, neighbor, stale
       State: Oper; Msgs sent/rcvd: 25/26

R1#clear mpls ldp nei 2.2.2.2

R1#show mpls ldp neighbor password
    Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0
       TCP connection: 2.2.2.2.63845 - 1.1.1.1.646
       Password: not required, neighbor, in use
       State: Oper; Msgs sent/rcvd: 25/24
```

```
R3#show mpls ldp neighbor detail | in Peer LDP|Password
    Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 3.3.3.3:0
       Password: not required, fallback, stale
```

```
    Peer LDP Ident: 6.6.6.6:0; Local LDP Ident 3.3.3.3:0
       Password: not required, fallback, stale
    Peer LDP Ident: 4.4.4.4:0; Local LDP Ident 3.3.3.3:0
       Password: not required, option 1 (KC34), in use
```

BGP

Russo – Lab 1 ECMP

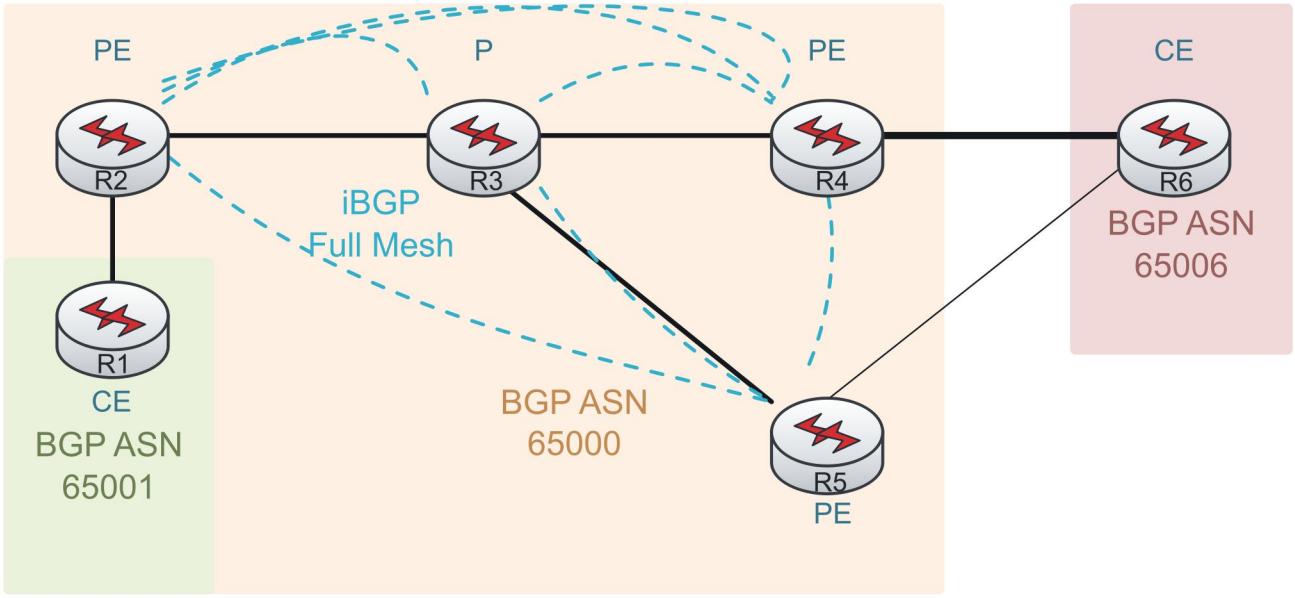


Figure 0-4 BGP Russo Lab 1 ECMP

Load Config

```
# R1 – R6
config replace unix:labconfigs/bgp.russo-lab1.cfg force
```

The idea behind these labs is that **R6** is advertising **192.0.2.6/32** and **2001:db8::6/128**. It is multi-homed to **R4** and **R5**. We will use various techniques to achieve load balancing and failover.

This first lab requires you to configure **ECMP**.

Currently **R2** and **R3** are using **R4** for **egress** due to its lower **RID**. R6 is using R4 for this reason as well.

Configure the fewest number of routers so that traffic from R1 to R6 is load balanced through R4 and R5, and return traffic from R6 is likewise load balanced.

Answer

```
# R3
router bgp 65000
 address-family ipv4
  maximum-paths ibgp 2
 address-family ipv6
```

```
  maximum-paths ibgp 2

# R6
router bgp 65006
 address-family ipv4
  maximum-paths 2
 address-family ipv6
  maximum-paths 2
```

Explanation

We do not need to enable iBGP multipath on **R2**, it will always use R3 to get to R4 and R5, so enabling multipath does not technically achieve anything. It can be done, but the result is the same with or without it.

```
R3#show bgp 192.0.2.6
% Command accepted but obsolete, unreleased or unsupported; see documentation.
BGP routing table entry for 192.0.2.6/32, version 3
Paths: (2 available, best #2, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65006
    10.0.0.5 (metric 10) from 10.0.0.5 (10.0.0.5)
      Origin IGP, metric 0, localpref 100, valid, internal
      rx pathid: 0, tx pathid: 0
      Updated on Oct 20 2026 06:16:26 UTC
  Refresh Epoch 1
  65006
    10.0.0.4 (metric 10) from 10.0.0.4 (10.0.0.4)
      Origin IGP, metric 0, localpref 100, valid, internal, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 20 2026 06:16:25 UTC

R3#
router bgp 65000
 address-family ipv4
  maximum-paths ibgp 2

R3#show ip bgp 192.0.2.6
BGP routing table entry for 192.0.2.6/32, version 4
Paths: (2 available, best #2, table default)
Multipath: iBGP
  Not advertised to any peer
  Refresh Epoch 1
  65006
    10.0.0.5 (metric 10) from 10.0.0.5 (10.0.0.5)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath(oldest)
      rx pathid: 0, tx pathid: 0
      Updated on Oct 20 2026 06:16:26 UTC
  Refresh Epoch 1
  65006
    10.0.0.4 (metric 10) from 10.0.0.4 (10.0.0.4)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 20 2026 06:16:25 UTC

R3#show ip route 192.0.2.6
Routing entry for 192.0.2.6/32
  Known via "bgp 65000", distance 200, metric 0
```

```
  Tag 65006, type internal
  Last update from 10.0.0.4 00:00:23 ago
  Routing Descriptor Blocks:
  * 10.0.0.5, from 10.0.0.5, 00:00:23 ago
      opaque_ptr 0x7F2E06A9E920
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65006
      MPLS label: none
    10.0.0.4, from 10.0.0.4, 00:00:23 ago
      opaque_ptr 0x7F2E06A9E920
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65006
      MPLS label: none
```

Russo – Lab 2 UCMP

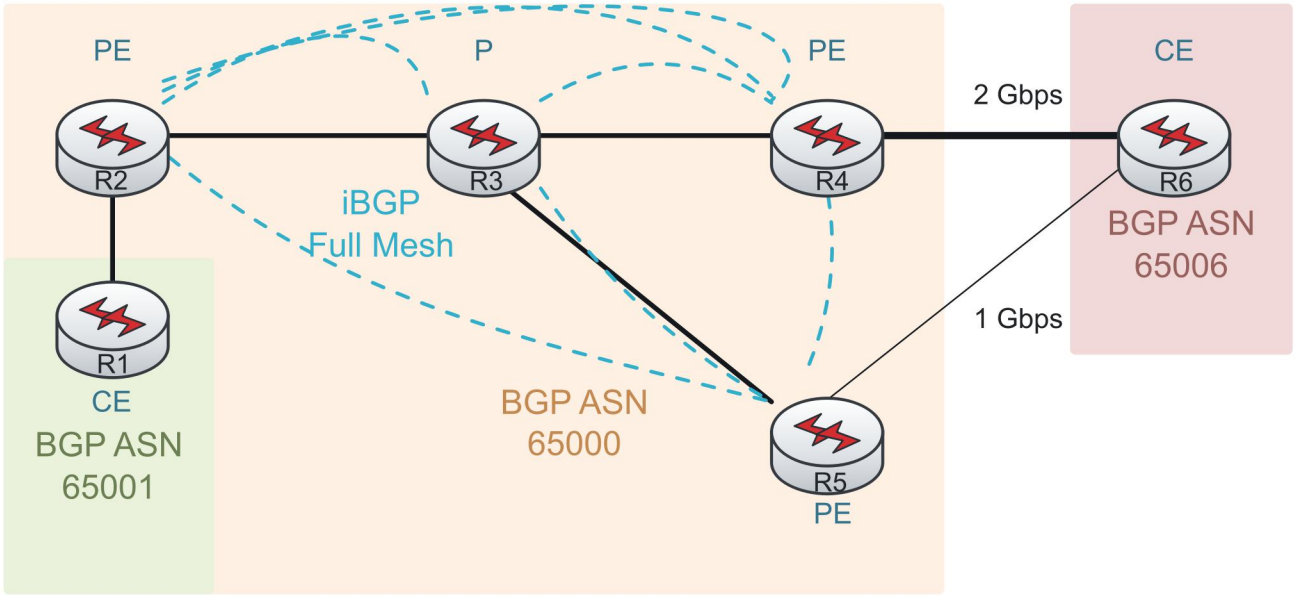


Figure 0-5 BGP Russo Lab 2 UCMP

Load Config

```
# R1 – R6
config replace unix:labconfigs/bgp.russo-lab2.cfg force
```

The bandwidth of the R4-R6 and R5-R6 link has been changed, matching the diagram above.

- Configure UCMP so that **R3** will send twice as much traffic to **R4**.
- Additionally, configure **R6** to send twice as much traffic to **R4**.
- Only do this for **IPv4 traffic**. (It doesn't work for IPv6).

Answer

```
# R3
router bgp 65000
 add ipv4
  maximum-paths ibgp 2
  bgp dmzlink-bw

# R4
router bgp 65000
 template peer-policy IBGP_V4V6
  send-community both
 exit-peer-policy
 add ipv4
  bgp dmzlink-bw
  neighbor 10.4.6.6 dmzlink-bw

# R5
```

```
router bgp 65000
 template peer-policy IBGP_V4V6
  send-community both
 exit-peer-policy
 add ipv4
  bgp dmzlink-bw
  neighbor 10.5.6.6 dmzlink-bw

# R6
router bgp 65006
 add ipv4
  bgp dmzlink-bw
  neighbor 10.4.6.4 dmzlink-bw
  neighbor 10.5.6.5 dmzlink-bw
  maximum-paths 2
```

Explanation

```
R3#show ip bgp 192.0.2.6
BGP routing table entry for 192.0.2.6/32, version 7
Paths: (2 available, best #2, table default)
Multipath: iBGP
  Not advertised to any peer
  Refresh Epoch 9
  65006
    10.0.0.5 (metric 10) from 10.0.0.5 (10.0.0.5)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath(oldest)
      DMZ-Link Bw 125000 kbytes
      rx pathid: 0, tx pathid: 0
      Updated on Oct 20 2026 06:39:18 UTC
  Refresh Epoch 9
  65006
    10.0.0.4 (metric 10) from 10.0.0.4 (10.0.0.4)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath, best
      DMZ-Link Bw 250000 kbytes
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 20 2026 06:39:08 UTC

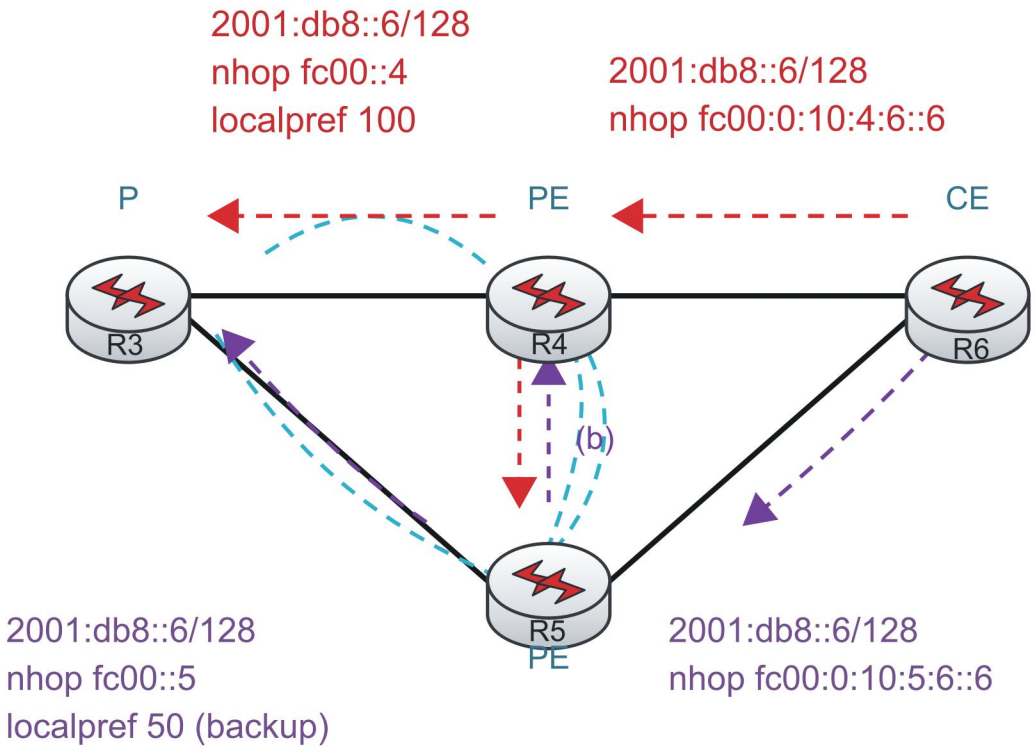
R3#show ip route 192.0.2.6
Routing entry for 192.0.2.6/32
  Known via "bgp 65000", distance 200, metric 0
  Tag 65006, type internal
  Last update from 10.0.0.4 00:00:12 ago
  Routing Descriptor Blocks:
  * 10.0.0.5, from 10.0.0.5, 00:00:12 ago
      opaque_ptr 0x7F2E06A9E920
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65006
      MPLS label: none
    10.0.0.4, from 10.0.0.4, 00:00:12 ago
      opaque_ptr 0x7F2E06A9E920
      Route metric is 0, traffic share count is 2
      AS Hops 1
      Route tag 65006
      MPLS label: none

R6#show ip bgp 192.0.2.1
BGP routing table entry for 192.0.2.1/32, version 5
```

```
Paths: (2 available, best #2, table default)
Multipath: eBGP
  Advertised to update-groups:
     1
  Refresh Epoch 8
  65000 65001
    10.5.6.5 from 10.5.6.5 (10.0.0.5)
      Origin IGP, localpref 100, valid, external, multipath(oldest)
      DMZ-Link Bw 125000 kbytes
      rx pathid: 0, tx pathid: 0
      Updated on Oct 20 2026 06:45:26 UTC
  Refresh Epoch 7
  65000 65001
    10.4.6.4 from 10.4.6.4 (10.0.0.4)
      Origin IGP, localpref 100, valid, external, multipath, best
      DMZ-Link Bw 250000 kbytes
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 20 2026 06:45:26 UTC

R6#show ip route 192.0.2.1
Routing entry for 192.0.2.1/32
  Known via "bgp 65006", distance 20, metric 0
  Tag 65000, type external
  Last update from 10.4.6.4 00:01:02 ago
  Routing Descriptor Blocks:
  * 10.5.6.5, from 10.5.6.5, 00:01:02 ago
      opaque_ptr 0x7F1FD17E3530
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65000
      MPLS label: none
    10.4.6.4, from 10.4.6.4, 00:01:02 ago
      opaque_ptr 0x7F1FD17E3530
      Route metric is 0, traffic share count is 2
      AS Hops 2
      Route tag 65000
      MPLS label: none
```

Russo – Lab 3 Backup Path



Load Config

```
# R1 – R6
config replace unix:labconfigs/bgp.russo-lab3.cfg force
```

R5 is setting **Local Preference** to **50** on routes received from **R6**.

Configure the core so that **R4** is always used as the **primary path**, but if that path is lost, the core will **immediately re-route** to using **R5**.

Answer

```
#R5
router bgp 65000
 add ipv4
  bgp advertise-best-external
 add ipv6
  bgp advertise-best-external

#R2, R3, R4
router bgp 65000
 add ipv4
  bgp additional-paths install
 add ipv6
  bgp additional-paths install
```

Explanation

Before making any changes, notice that **R5** is **hiding** its **external path**. This is because its iBGP path via R4 is better than its eBGP path due to local preference. On R3, we only see R4's path to R6's prefixes.

```
# r5
R5#show ip bgp
     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.0.2.1/32     10.0.0.2                 0    100      0 65001 i
 *>i  192.0.2.6/32     10.0.0.4                 0    100      0 65006 i
 *                     10.5.6.6                 0     50      0 65006 i


R5#show bgp ipv6 unicast
     Network          Next Hop            Metric LocPrf Weight Path
 *>i  2001:DB8::1/128  FC00::2                  0    100      0 65001 i
 *>i  2001:DB8::6/128  FC00::4                  0    100      0 65006 i
 *                     FC00:10:5:6::6           0     50      0 65006 i
```

First we must configure R5 to advertise its external path. We can either use Add Path, or simply use the **advertise-best-external** feature. This allows **R5** to advertise **its eBGP path** even when the **iBGP path is best**, and does not require Add Path.

```
# r5
router bgp 65000
 add ipv4
  bgp advertise-best-external
 add ipv6
  bgp advertise-best-external


R5#show bgp ipv4 unicast
BGP table version is 10, local router ID is 10.0.0.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.0.2.1/32     10.0.0.2                 0    100      0 65001 i
 *>i  192.0.2.6/32     10.0.0.4                 0    100      0 65006 i
 *b x                  10.5.6.6                 0     50      0 65006 i

R5#show bgp ipv6 unicast
BGP table version is 15, local router ID is 10.0.0.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
```

```
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  2001:DB8::1/128  FC00::2                  0    100      0 65001 i
 *>i  2001:DB8::6/128  FC00::4                  0    100      0 65006 i
 *b x                  FC00:10:5:6::6           0     50      0 65006 i


R3#show bgp ipv6 uni
R3#show bgp ipv6 unicast | begin Network
     Network          Next Hop            Metric LocPrf Weight Path
 *>i  2001:DB8::1/128  FC00::2                  0    100      0 65001 i
 * i  2001:DB8::6/128  FC00::5                  0     50      0 65006 i
 *>i                   FC00::4                  0    100      0 65006 i
R3#show bgp ipv4 unicast | begin Network
     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.0.2.1/32     10.0.0.2                 0    100      0 65001 i
 * i  192.0.2.6/32     10.0.0.5                 0     50      0 65006 i
 *>i                   10.0.0.4                 0    100      0 65006 i
R3#show bgp ipv4 unicast 192.0.2.6
BGP routing table entry for 192.0.2.6/32, version 11
Paths: (2 available, best #2, table default)
  Flag: 0x100
  Not advertised to any peer
  Refresh Epoch 1
  65006
    10.0.0.5 (metric 10) from 10.0.0.5 (10.0.0.5)
      Origin IGP, metric 0, localpref 50, valid, internal
      rx pathid: 0, tx pathid: 0
      Updated on Oct 20 2026 09:10:33 UTC
  Refresh Epoch 2
  65006
    10.0.0.4 (metric 10) from 10.0.0.4 (10.0.0.4)
      Origin IGP, metric 0, localpref 100, valid, internal, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 20 2026 09:10:30 UTC
```

We need to tell **R2**, **R3**, and **R4** to install **additional paths** as **repair paths** in the FIB.

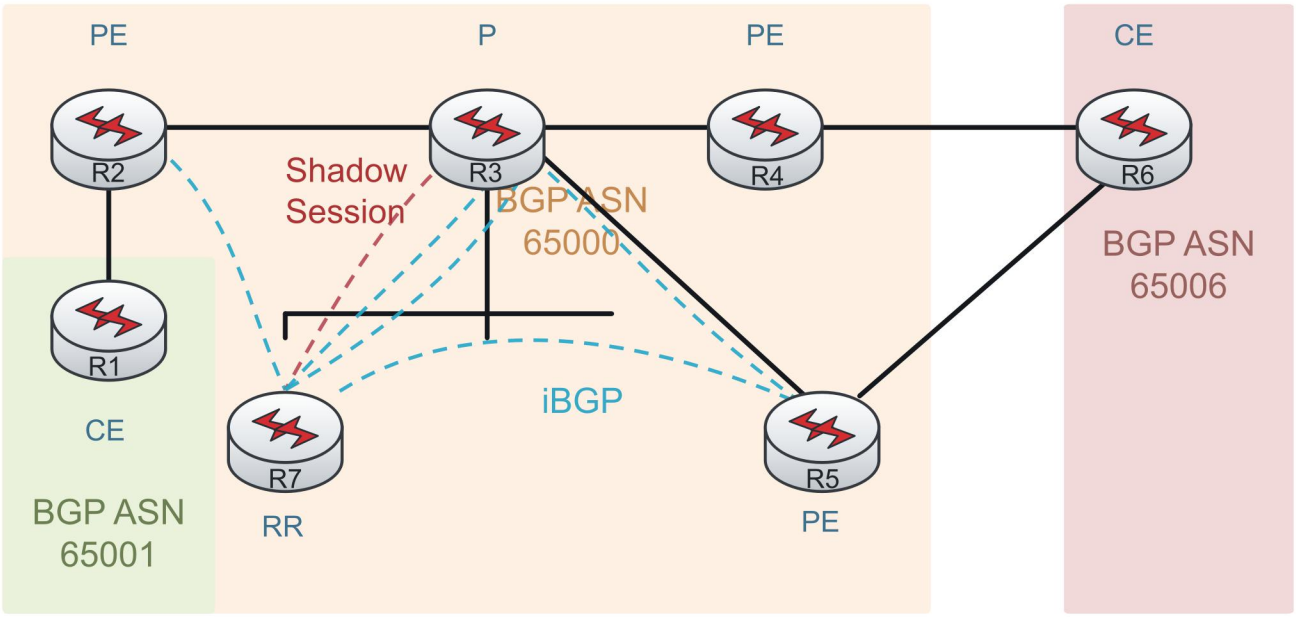
```
# r3, r2, r4
router bgp 65000
 add ipv4
  bgp additional-paths install
 add ipv6
  bgp additional-paths install

R3#show bgp ipv4 unicast 192.0.2.6/32
BGP routing table entry for 192.0.2.6/32, version 13
Paths: (2 available, best #2, table default)
  Additional-path-install
  Not advertised to any peer
  Refresh Epoch 2
  65006
    10.0.0.5 (metric 10) from 10.0.0.5 (10.0.0.5)
      Origin IGP, metric 0, localpref 50, valid, internal, backup/repair
      rx pathid: 0, tx pathid: 0
      Updated on Oct 20 2026 09:24:43 UTC
  Refresh Epoch 3
  65006
    10.0.0.4 (metric 10) from 10.0.0.4 (10.0.0.4)
      Origin IGP, metric 0, localpref 100, valid, internal, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 20 2026 09:24:43 UTC
```

```
R3#show ip cef 192.0.2.6
192.0.2.6/32
  nexthop 10.3.4.4 Ethernet0/1.34

R3#show ip cef 192.0.2.6 detail
192.0.2.6/32, epoch 0, flags [rib only nolabel, rib defined all labels]
  recursive via 10.0.0.4
    nexthop 10.3.4.4 Ethernet0/1.34
  recursive via 10.0.0.5, repair
    nexthop 10.3.5.5 Ethernet0/1.35
```

Russo – Lab 4 Shadow Session



Load Config

```
# R1 – R7
config replace unix:labconfigs/bgp.russo-lab4.cfg force
```

R7 has been introduced as the **RR**. This causes the path to **R6's loopbacks** via R5 to be hidden again.

Create a **shadow session** between **R3** and **R7** so that **R3** learns both paths (via R4 and via R5). Enable **multipath** on **R3**. New loopbacks (Lo1) have been preconfigured on R3 and R7 for this shadow session.

- R3 Lo1 = 10.0.0.33, fc00::33
- R7 Lo1 = 10.0.0.77, fc00::77

Answer

```
# R3
router bgp 65000
 template peer-session IBGP_SESS_SHADOW
  remote-as 65000
  update-source Loopback1
 !
 neighbor 10.0.0.77 inherit peer-session IBGP_SESS_SHADOW
 neighbor FC00::77 inherit peer-session IBGP_SESS_SHADOW
 !
 address-family ipv4
  neighbor 10.0.0.77 activate
```

```
  maximum-paths ibgp 2
 !
 address-family ipv6
  maximum-paths ibgp 2
  neighbor FC00::77 activate

# R7
router bgp 65000
 !
 template peer-session IBGP_SESS_SHADOW
  remote-as 65000
  update-source Loopback1
 !
 neighbor 10.0.0.33 inherit peer-session IBGP_SESS_SHADOW
 neighbor FC00::33 inherit peer-session IBGP_SESS_SHADOW
 !
 address-family ipv4
  bgp additional-paths select backup
  neighbor 10.0.0.33 activate
  neighbor 10.0.0.33 inherit peer-policy IBGP_V4V6
  neighbor 10.0.0.33 advertise diverse-path backup
 exit-address-family
 !
 address-family ipv6
  bgp additional-paths select backup
  neighbor FC00::33 activate
  neighbor FC00::33 inherit peer-policy IBGP_V4V6
  neighbor FC00::33 advertise diverse-path backup
 exit-address-family
```

Explanation

Before we make the change, notice that R7 is hiding the backup path via R5. By default, the **RR only sends its bestpath** to all RR clients.

The RR advertises the "**diverse-path backup**" path on the shadow session, *in addition to* the bestpath. This does not require Add Path to be configured and negotiated between peers. It is similar to **advertise-best-external** in a way.

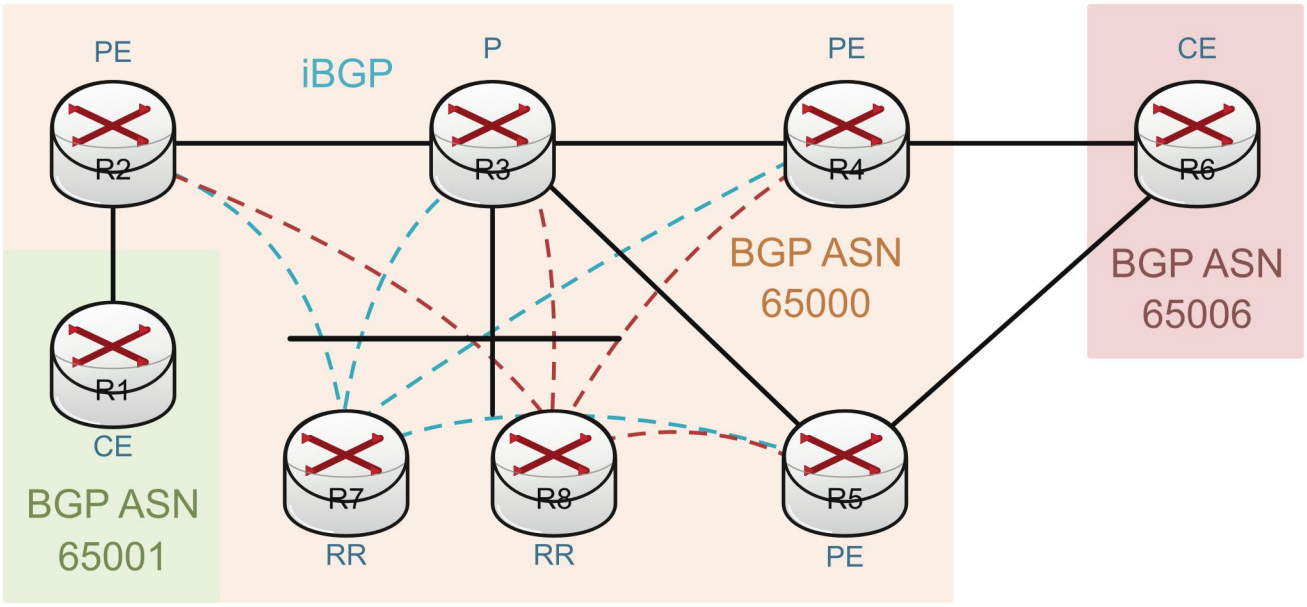
Also note that R7 does not need to locally install the backup path as a fast repair route, as it will not be in the data path.

```
R7#show bgp ipv4 unicast neighbors 10.0.0.33 advertised-routes
BGP table version is 5, local router ID is 10.0.0.7
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.0.2.1/32     10.0.0.2                 0    100      0 65001 i
 *bia 192.0.2.6/32     10.0.0.5                 0    100      0 65006 i
```

```
R3#show bgp ipv4 unicast 192.0.2.6
BGP routing table entry for 192.0.2.6/32, version 8
Paths: (2 available, best #2, table default)
Multipath: iBGP
  Not advertised to any peer
  Refresh Epoch 2
  65006
    10.0.0.5 (metric 10) from 10.0.0.77 (10.0.0.7)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath(oldest)
      Originator: 10.0.0.5, Cluster list: 10.0.0.7
      rx pathid: 0, tx pathid: 0
      Updated on Oct 24 2026 07:16:32 UTC
  Refresh Epoch 1
  65006
    10.0.0.4 (metric 10) from 10.0.0.7 (10.0.0.7)
      Origin IGP, metric 0, localpref 100, valid, internal, multipath, best
      Originator: 10.0.0.4, Cluster list: 10.0.0.7
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 24 2026 07:16:32 UTC
```

Russo – Lab 5 Shadow RR



Load Config

```
# R1 – R8
config replace unix:labconfigs/bgp.russo-lab5.cfg force
```

R8 has been introduced as a **second RR**.

- Configure this as a **shadow RR** so that **R8** advertises the backup path **via R5** to all **route-reflector clients**.
    - The second RR session has already been configured on all routers within 65000.
- **R3** is already configured for **iBGP multipath**.
- Ensure that the **RRs** make the **same bestpath decision**, no matter where they are placed in the IGP topology.

Answer

```
# R1 – R8
#R7
router bgp 65000
 add ipv4
  bgp bestpath igp-metric ignore
 add ipv6
  bgp bestpath igp-metric ignore

#R8
router bgp 65000
 !
```

```
 add ipv4
  bgp bestpath igp-metric ignore
  bgp additional-paths select backup
 add ipv6
  bgp bestpath igp-metric ignore
  bgp additional-paths select backup
 !
 template peer-policy IBGP_V4V6
  advertise diverse-path backup
 exit-peer-policy
```

Explanation

To guarantee that the two RRs (R7 and R8) will choose the same primary bestpath no matter where they are in the topology, we must **disable the IGP metric** comparison in the **bestpath** decision process. This ensures that the RRs always select the same bestpath.

```
#R7, R8
router bgp 65000
 add ipv4
  bgp bestpath igp-metric ignore
 add ipv6
  bgp bestpath igp-metric ignore
```

Next, on **R8**, we must **only advertise** the **diverse backup** path to all RR clients. We select the backup path, and then advertise this to peers using the template peer-policy. Note that the backup paths must be first selected before you can apply the advertisement of the backup path in the template. You may need to **clear ip bgp *** after applying the config.

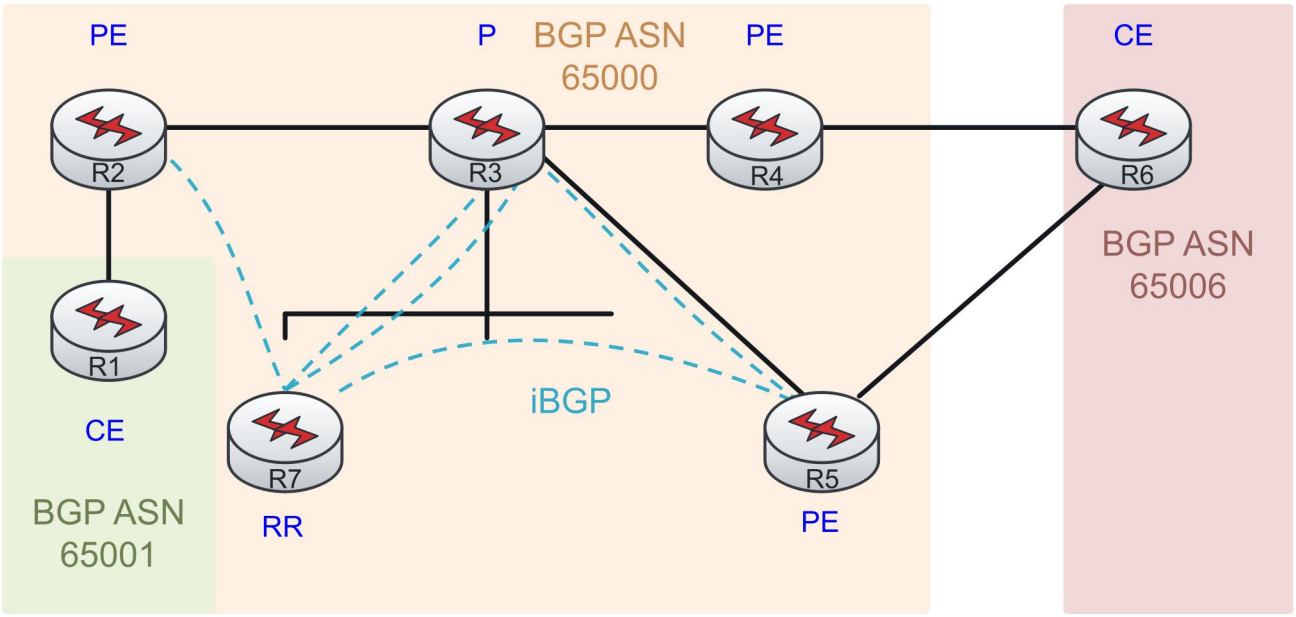
```
#R8
router bgp 65000
 !
 add ipv4
  bgp additional-paths select backup
 add ipv6
  bgp additional-paths select backup
 !
 template peer-policy IBGP_V4V6
  advertise diverse-path backup
 exit-peer-policy


R8#show bgp ipv4 unicast neighbors 10.0.0.3 advertised-routes
BGP table version is 7, local router ID is 10.0.0.8
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
```

```
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>i  192.0.2.1/32     10.0.0.2                 0    100      0 65001 i
 *bia 192.0.2.6/32     10.0.0.5                 0    100      0 65006 i
```

Russo – Lab 6 RR with Add-Path



Load Config

```
# R1 – R6
config replace unix:labconfigs/bgp.russo-lab6.cfg force
```

R8 has been removed as an **RR**. **R5** is setting **LP=50** on routes received from R6, so that **R4** is the **preferred egress** point in the core.

Configure **add path** so that **R7** will reflect both a **primary** and **backup** path to all iBGP route reflector clients. On the **clients**, configure **PIC** so that the backup path is pre-installed as a repair route.

Answer

```
# AS 65000
router bgp 65000
 add ipv4
  bgp additional-paths send receive
  bgp additional-paths select best 2

# R2,R3,R4,R5
router bgp 65000
 add ipv4
  bgp additional-paths install

# R5 & R7
router bgp 65000
 template peer-policy IBGP_V4V6
  advertise additional-path best 2
```

Explanation

First we must enable the **Add Path capability** on all routers. This is a negotiated capability in the OPEN message, and once configured, the router will immediately tear down the existing session.

```
# AS 65000
router bgp 65000
 add ipv4
  bgp additional-paths send receive
 add ipv6
  bgp additional-paths send receive

R7#show bgp ipv4 unicast neighbors | in Additional
  Additional Paths send capability: advertised and received
  Additional Paths receive capability: advertised and received
  Additional Paths send capability: advertised and received
  Additional Paths receive capability: advertised and received
  Additional Paths send capability: advertised and received
  Additional Paths receive capability: advertised and received
  Additional Paths send capability: advertised and received
  Additional Paths receive capability: advertised and received
```

On R7, verify that it is receiving two paths for R6's loopbacks. Each path is kept separate using a unique pathid (0x0 and 0x1). The paths via R5 have a lower LP, so these are the "best2" paths.

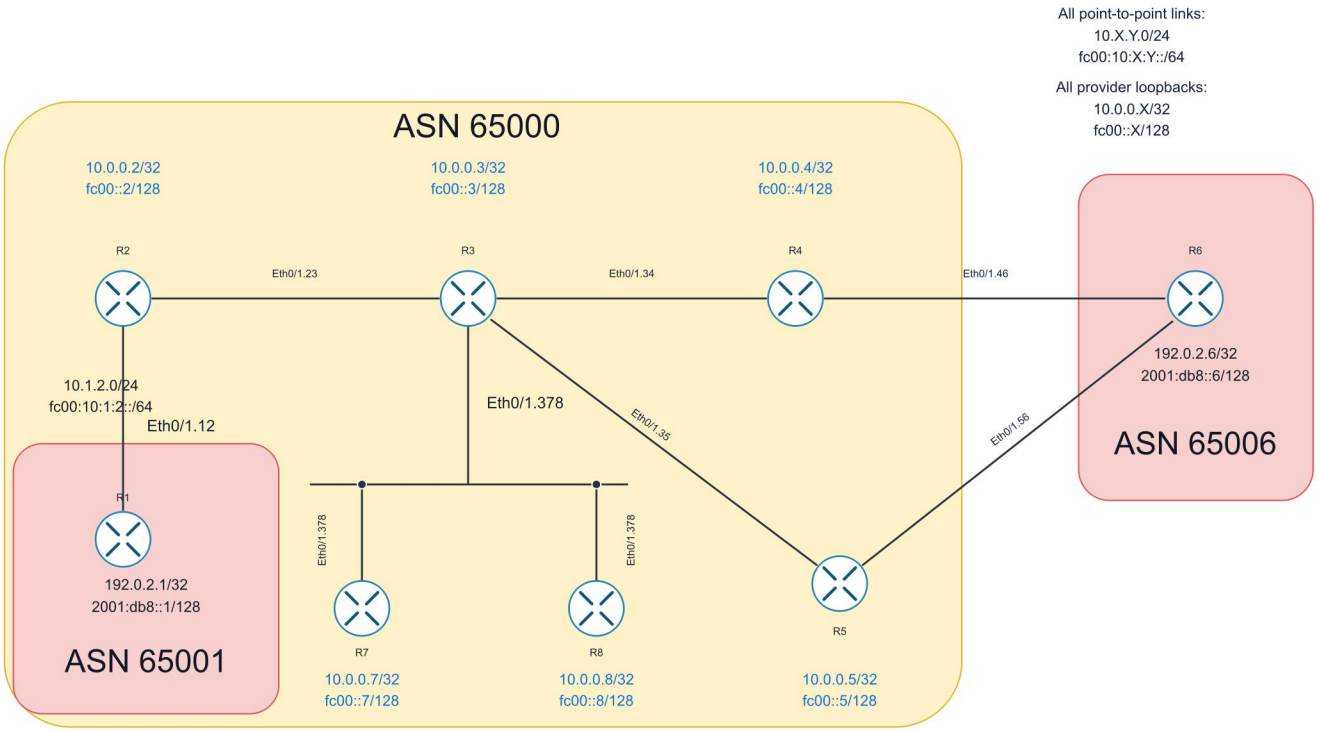
```
R7#show bgp ipv4 unicast 192.0.2.6
BGP routing table entry for 192.0.2.6/32, version 5
Paths: (2 available, best #2, table default)
  Path advertised to update-groups:
     7
  Refresh Epoch 8
  65006, (Received from a RR-client)
    10.0.0.5 (metric 20) from 10.0.0.5 (10.0.0.5)
      Origin IGP, metric 0, localpref 50, valid, internal, best2
      rx pathid: 0x1, tx pathid: 0x1
      Updated on Nov 2 2026 01:08:57 UTC
  Path advertised to update-groups:
     7
  Refresh Epoch 8
  65006, (Received from a RR-client)
    10.0.0.4 (metric 20) from 10.0.0.4 (10.0.0.4)
      Origin IGP, metric 0, localpref 100, valid, internal, best
      rx pathid: 0x0, tx pathid: 0x0
      Updated on Nov 2 2026 01:08:57 UTC
```

On each **RR client**, it will receive both paths from the RR. The path IDs are used by the Add Path feature to ensure that a second Update for the same prefix does not implicitly withdraw the first Update. The paths via R5 are used as repair routes.

```
R3#show bgp ipv4 unicast 192.0.2.6
BGP routing table entry for 192.0.2.6/32, version 5
```

```
Paths: (2 available, best #2, table default)
  Additional-path-install
  Path not advertised to any peer
  Refresh Epoch 8
  65006
    10.0.0.5 (metric 10) from 10.0.0.7 (10.0.0.7)
      Origin IGP, metric 0, localpref 50, valid, internal, backup/repair, best2
      Originator: 10.0.0.5, Cluster list: 10.0.0.7
      rx pathid: 0x1, tx pathid: 0x1
      Updated on Nov 2 2026 01:08:56 UTC
  Path not advertised to any peer
  Refresh Epoch 8
  65006
    10.0.0.4 (metric 10) from 10.0.0.7 (10.0.0.7)
      Origin IGP, metric 0, localpref 100, valid, internal, best
      Originator: 10.0.0.4, Cluster list: 10.0.0.7
      rx pathid: 0x0, tx pathid: 0x0
      Updated on Nov 2 2026 01:08:56 UTC
```

Russo – Conditional Advertisement



Load Config

```
# R1 – R6
config replace unix:labconfigs/bgp.russo-lab1.cfg force
```

- Configure **R4** and **R5** to advertise an IPv4 and IPv6 **default route** to **R6**.
- **R6** should prefer the **default routes via R4**.

– Configure **R6** to only advertise its IPv4/IPv6 loopback addresses to **R5** if the **default route** via R4 is gone.

Answer

```
# R4

router bgp 65000
 address-family ipv4
  neighbor 10.4.6.6 default-originate
 address-family ipv6
  neighbor FC00:10:4:6::6 default-originate

# R5

router bgp 65000
 address-family ipv4
  neighbor 10.5.6.6 default-originate
 address-family ipv6
  neighbor FC00:10:5:6::6 default-originate

# R6
route-map FROM_R4 permit 10
 set community 65000:4

ip prefix-list R4_DEF seq 5 permit 0.0.0.0/0
ip prefix-list lo0 seq 5 permit 192.0.2.6/32

route-map lo0 permit 10
 match ip address prefix-list lo0

ipv6 prefix-list R4_DEF6 seq 5 permit ::/0
ipv6 prefix-list lo0v6 seq 5 permit 2001:DB8::6/128

route-map lo0v6 permit 10
 match ipv6 address prefix-list lo0v6

ip community-list standard R4_DEFAULT permit 65000:4

route-map R4_DEF permit 10
 match ip address prefix-list R4_DEF
 match community R4_DEFAULT

route-map R4_DEF6 permit 10
 match community R4_DEFAULT
 match ipv6 address prefix-list R4_DEF6

router bgp 65006
 address-family ipv4
  neighbor 10.4.6.4 route-map FROM_R4 in
  neighbor 10.5.6.5 advertise-map lo0 non-exist-map R4_DEF
 exit-address-family
 !
 address-family ipv6
  neighbor FC00:10:4:6::4 route-map FROM_R4 in
  neighbor FC00:10:5:6::5 advertise-map lo0v6 non-exist-map R4_DEF6
```

Explanation

R6 must use **prefix lists** to identify the **default routes** and the **local routes** to advertise to **R5**. Additionally, we need to match **only R4**'s default route. We assign a **community** to all routes received from **R4**, and match on this **community**.

```
# R6

R6#show bgp ipv4 unicast 0.0.0.0
BGP routing table entry for 0.0.0.0/0, version 6
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     7
  Refresh Epoch 12
  65000
    10.4.6.4 from 10.4.6.4 (10.0.0.4)
      Origin IGP, localpref 100, valid, external
      Community: 4259840004
      rx pathid: 0, tx pathid: 0
      Updated on Oct 31 2026 09:15:55 UTC
  Refresh Epoch 6
  65000
    10.5.6.5 from 10.5.6.5 (10.0.0.5)
      Origin IGP, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 31 2026 09:15:56 UTC
R6#show bgp ipv4 unicast neighbors 10.5.6.5 | in Advert
  Index 8, Advertise bit 1
  Condition-map R4_DEF, Advertise-map lo0, status: Withdraw

# R4
router bgp 65000
 add ipv4
  no neighbor 10.4.6.6 default-originate

# R6

R6#show bgp ipv4 unicast 0.0.0.0
BGP routing table entry for 0.0.0.0/0, version 6
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     7
  Refresh Epoch 6
  65000
    10.5.6.5 from 10.5.6.5 (10.0.0.5)
      Origin IGP, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
      Updated on Oct 31 2026 09:15:56 UTC

R6#show bgp ipv4 unicast neighbors 10.5.6.5 | in Advert
  Index 8, Advertise bit 1
  Condition-map R4_DEF, Advertise-map lo0, status: Advertise
```